

COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./DUE DATE PD 98-1650 08/15/18		☐ Special Exception to Deadline Date Policy		FOR NSF USE ONLY NSF PROPOSAL NUMBER 1851510	
FOR CONSIDERATION BY NSF ORGANIZATION UNIT(S) (Indicate the most specific unit known, i.e. program, division, etc.) OCE - BIOLOGICAL OCEANOGRAPHY					
DATE RECEIVED	NUMBER OF COPIES	DIVISION ASSIGNED	FUND CODE	DUNS# (Data Universal Numbering System)	FILE LOCATION
08/15/2018	2	06040000 OCE	1650	094878394	08/16/2018 2:29pm S
EMPLOYER IDENTIFICATION NUMBER (EIN) OR TAXPAYER IDENTIFICATION NUMBER (TIN) 956006145		SHOW PREVIOUS AWARD NO. IF THIS IS ☐ A RENEWAL ☐ AN ACCOMPLISHMENT-BASED RENEWAL		IS THIS PROPOSAL BEING SUBMITTED TO ANOTHER FEDERAL AGENCY? YES ☐ NO ☒ IF YES, LIST ACRONYM(S)	
NAME OF ORGANIZATION TO WHICH AWARD SHOULD BE MADE University of California-Santa Barbara		ADDRESS OF Awardee ORGANIZATION, INCLUDING 9 DIGIT ZIP CODE Office of Research Rm 3227 Cheadle Hall Santa Barbara, CA 93106-2050			
AWARDEE ORGANIZATION CODE (IF KNOWN) 0013201000					
NAME OF PRIMARY PLACE OF PERF University of California-Santa Barbara		ADDRESS OF PRIMARY PLACE OF PERF, INCLUDING 9 DIGIT ZIP CODE University of California-Santa Barbara Santa Barbara ,CA ,931066150 ,US.			
IS AWARDEE ORGANIZATION (Check All That Apply)		☐ SMALL BUSINESS ☐ FOR-PROFIT ORGANIZATION		☐ MINORITY BUSINESS ☐ WOMAN-OWNED BUSINESS ☐ IF THIS IS A PRELIMINARY PROPOSAL THEN CHECK HERE	
TITLE OF PROPOSED PROJECT COLLABORATIVE RESEARCH: Dynamic Marine Landscapes: Feedbacks and spatial patterns of corals and their associated fishes					
REQUESTED AMOUNT \$ 651,134	PROPOSED DURATION (1-60 MONTHS) 36 months	REQUESTED STARTING DATE 05/01/19	SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE		
THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW ☐ BEGINNING INVESTIGATOR ☐ DISCLOSURE OF LOBBYING ACTIVITIES ☐ PROPRIETARY & PRIVILEGED INFORMATION ☐ HISTORIC PLACES ☒ VERTEBRATE ANIMALS IACUC App. Date 06/23/17 PHS Animal Welfare Assurance Number FWA00006361 ☒ TYPE OF PROPOSAL Research					
☐ HUMAN SUBJECTS Human Subjects Assurance Number _____ Exemption Subsection _____ or IRB App. Date _____ ☒ INTERNATIONAL ACTIVITIES: COUNTRY/COUNTRIES INVOLVED FP ☒ COLLABORATIVE STATUS A collaborative proposal from multiple organizations (PAPPG II.D.3.b)					
PI/PD DEPARTMENT Biology		PI/PD POSTAL ADDRESS Office of Research Rm 3227 Cheadle Hall SANTA BARBARA, CA 931062050 United States			
PI/PD FAX NUMBER 805-893-5885					
NAMES (TYPED)	High Degree	Yr of Degree	Telephone Number	Email Address	
PI/PD NAME Adrian C Stier	PhD	2012	805-893-4188	adrian.stier@lifesci.ucsb.edu	
CO-PI/PD					
CO-PI/PD					
CO-PI/PD					
CO-PI/PD					

CERTIFICATION PAGE

Certification for Authorized Organizational Representative (or Equivalent) or Individual Applicant

By electronically signing and submitting this proposal, the Authorized Organizational Representative (AOR) or Individual Applicant is: (1) certifying that statements made herein are true and complete to the best of his/her knowledge; and (2) agreeing to accept the obligation to comply with NSF award terms and conditions if an award is made as a result of this application. Further, the applicant is hereby providing certifications regarding conflict of interest (when applicable), drug-free workplace, debarment and suspension, lobbying activities (see below), nondiscrimination, flood hazard insurance (when applicable), responsible conduct of research, organizational support, Federal tax obligations, unpaid Federal tax liability, and criminal convictions as set forth in the NSF Proposal & Award Policies & Procedures Guide (PAPPG). Willful provision of false information in this application and its supporting documents or in reports required under an ensuing award is a criminal offense (U.S. Code, Title 18, Section 1001).

Certification Regarding Conflict of Interest

The AOR is required to complete certifications stating that the organization has implemented and is enforcing a written policy on conflicts of interest (COI), consistent with the provisions of PAPPG Chapter IX.A.; that, to the best of his/her knowledge, all financial disclosures required by the conflict of interest policy were made; and that conflicts of interest, if any, were, or prior to the organization's expenditure of any funds under the award, will be, satisfactorily managed, reduced or eliminated in accordance with the organization's conflict of interest policy. Conflicts that cannot be satisfactorily managed, reduced or eliminated and research that proceeds without the imposition of conditions or restrictions when a conflict of interest exists, must be disclosed to NSF via use of the Notifications and Requests Module in FastLane.

Drug Free Work Place Certification

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent), is providing the Drug Free Work Place Certification contained in Exhibit II-3 of the Proposal & Award Policies & Procedures Guide.

Debarment and Suspension Certification

(If answer "yes", please provide explanation.)

Is the organization or its principals presently debarred, suspended, proposed for debarment, declared ineligible, or voluntarily excluded from covered transactions by any Federal department or agency?

Yes ☐

No ☒

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) or Individual Applicant is providing the Debarment and Suspension Certification contained in Exhibit II-4 of the Proposal & Award Policies & Procedures Guide.

Certification Regarding Lobbying

This certification is required for an award of a Federal contract, grant, or cooperative agreement exceeding \$100,000 and for an award of a Federal loan or a commitment providing for the United States to insure or guarantee a loan exceeding \$150,000.

Certification for Contracts, Grants, Loans and Cooperative Agreements

The undersigned certifies, to the best of his or her knowledge and belief, that:

- (1) No Federal appropriated funds have been paid or will be paid, by or on behalf of the undersigned, to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with the awarding of any Federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.
- (2) If any funds other than Federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with this Federal contract, grant, loan, or cooperative agreement, the undersigned shall complete and submit Standard Form-LLL, "Disclosure of Lobbying Activities," in accordance with its instructions.
- (3) The undersigned shall require that the language of this certification be included in the award documents for all subawards at all tiers including subcontracts, subgrants, and contracts under grants, loans, and cooperative agreements and that all subrecipients shall certify and disclose accordingly.

This certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this certification is a prerequisite for making or entering into this transaction imposed by section 1352, Title 31, U.S. Code. Any person who fails to file the required certification shall be subject to a civil penalty of not less than \$10,000 and not more than \$100,000 for each such failure.

Certification Regarding Nondiscrimination

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is providing the Certification Regarding Nondiscrimination contained in Exhibit II-6 of the Proposal & Award Policies & Procedures Guide.

Certification Regarding Flood Hazard Insurance

Two sections of the National Flood Insurance Act of 1968 (42 USC §4012a and §4106) bar Federal agencies from giving financial assistance for acquisition or construction purposes in any area identified by the Federal Emergency Management Agency (FEMA) as having special flood hazards unless the:

- (1) community in which that area is located participates in the national flood insurance program; and
- (2) building (and any related equipment) is covered by adequate flood insurance.

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) or Individual Applicant located in FEMA-designated special flood hazard areas is certifying that adequate flood insurance has been or will be obtained in the following situations:

- (1) for NSF grants for the construction of a building or facility, regardless of the dollar amount of the grant; and
- (2) for other NSF grants when more than \$25,000 has been budgeted in the proposal for repair, alteration or improvement (construction) of a building or facility.

Certification Regarding Responsible Conduct of Research (RCR)

(This certification is not applicable to proposals for conferences, symposia, and workshops.)

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with the NSF Proposal & Award Policies & Procedures Guide, Chapter IX.B., the institution has a plan in place to provide appropriate training and oversight in the responsible and ethical conduct of research to undergraduates, graduate students and postdoctoral researchers who will be supported by NSF to conduct research. The AOR shall require that the language of this certification be included in any award documents for all subawards at all tiers.

CERTIFICATION PAGE - CONTINUED**Certification Regarding Organizational Support**

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that there is organizational support for the proposal as required by Section 526 of the America COMPETES Reauthorization Act of 2010. This support extends to the portion of the proposal developed to satisfy the Broader Impacts Review Criterion as well as the Intellectual Merit Review Criterion, and any additional review criteria specified in the solicitation. Organizational support will be made available, as described in the proposal, in order to address the broader impacts and intellectual merit activities to be undertaken.

Certification Regarding Federal Tax Obligations

When the proposal exceeds \$5,000,000, the Authorized Organizational Representative (or equivalent) is required to complete the following certification regarding Federal tax obligations. By electronically signing the Certification pages, the Authorized Organizational Representative is certifying that, to the best of their knowledge and belief, the proposing organization:

- (1) has filed all Federal tax returns required during the three years preceding this certification;
- (2) has not been convicted of a criminal offense under the Internal Revenue Code of 1986; and
- (3) has not, more than 90 days prior to this certification, been notified of any unpaid Federal tax assessment for which the liability remains unsatisfied, unless the assessment is the subject of an installment agreement or offer in compromise that has been approved by the Internal Revenue Service and is not in default, or the assessment is the subject of a non-frivolous administrative or judicial proceeding.

Certification Regarding Unpaid Federal Tax Liability

When the proposing organization is a corporation, the Authorized Organizational Representative (or equivalent) is required to complete the following certification regarding Federal Tax Liability:

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that the corporation has no unpaid Federal tax liability that has been assessed, for which all judicial and administrative remedies have been exhausted or lapsed, and that is not being paid in a timely manner pursuant to an agreement with the authority responsible for collecting the tax liability.

Certification Regarding Criminal Convictions

When the proposing organization is a corporation, the Authorized Organizational Representative (or equivalent) is required to complete the following certification regarding Criminal Convictions:

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that the corporation has not been convicted of a felony criminal violation under any Federal law within the 24 months preceding the date on which the certification is signed.

Certification Dual Use Research of Concern

By electronically signing the certification pages, the Authorized Organizational Representative is certifying that the organization will be or is in compliance with all aspects of the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern.



AUTHORIZED ORGANIZATIONAL REPRESENTATIVE		SIGNATURE		DATE
NAME Reese Raffo		Electronic Signature		Aug 15 2018 4:29PM
TELEPHONE NUMBER 805-893-2041	EMAIL ADDRESS raffo@research.ucsb.edu		FAX NUMBER	

COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./DUE DATE PD 98-1650 08/15/18		☐ Special Exception to Deadline Date Policy		FOR NSF USE ONLY NSF PROPOSAL NUMBER 1851032	
FOR CONSIDERATION BY NSF ORGANIZATION UNIT(S) (Indicate the most specific unit known, i.e. program, division, etc.) OCE - BIOLOGICAL OCEANOGRAPHY					
DATE RECEIVED	NUMBER OF COPIES	DIVISION ASSIGNED	FUND CODE	DUNS# (Data Universal Numbering System)	FILE LOCATION
08/14/2018	2	06040000 OCE	1650	004315578	08/16/2018 2:30pm S
EMPLOYER IDENTIFICATION NUMBER (EIN) OR TAXPAYER IDENTIFICATION NUMBER (TIN) 581353149		SHOW PREVIOUS AWARD NO. IF THIS IS ☐ A RENEWAL ☐ AN ACCOMPLISHMENT-BASED RENEWAL		IS THIS PROPOSAL BEING SUBMITTED TO ANOTHER FEDERAL AGENCY? YES ☐ NO ☒ IF YES, LIST ACRONYM(S)	
NAME OF ORGANIZATION TO WHICH AWARD SHOULD BE MADE University of Georgia Research Foundation Inc		ADDRESS OF AWARDEE ORGANIZATION, INCLUDING 9 DIGIT ZIP CODE University of Georgia Research Foundation Inc 310 East Campus Rd Athens, GA. 306021589			
AWARDEE ORGANIZATION CODE (IF KNOWN) 0015982001					
NAME OF PRIMARY PLACE OF PERF University of Georgia		ADDRESS OF PRIMARY PLACE OF PERF, INCLUDING 9 DIGIT ZIP CODE University of Georgia 140 E. Green St. Athens ,GA ,306021589 ,US.			
IS AWARDEE ORGANIZATION (Check All That Apply)		☐ SMALL BUSINESS ☐ FOR-PROFIT ORGANIZATION		☐ MINORITY BUSINESS ☐ WOMAN-OWNED BUSINESS ☐ IF THIS IS A PRELIMINARY PROPOSAL THEN CHECK HERE	
TITLE OF PROPOSED PROJECT Collaborative Research: Dynamic marine landscapes -- feedbacks and spatial patterns of corals and coral-associated fishes and invertebrates					
REQUESTED AMOUNT \$ 346,792	PROPOSED DURATION (1-60 MONTHS) 36 months	REQUESTED STARTING DATE 05/01/19	SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE		
THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW					
☐ BEGINNING INVESTIGATOR		☐ HUMAN SUBJECTS Human Subjects Assurance Number _____			
☐ DISCLOSURE OF LOBBYING ACTIVITIES		Exemption Subsection _____ or IRB App. Date _____			
☐ PROPRIETARY & PRIVILEGED INFORMATION		☒ INTERNATIONAL ACTIVITIES: COUNTRY/COUNTRIES INVOLVED			
☐ HISTORIC PLACES		FP			
☒ VERTEBRATE ANIMALS IACUC App. Date Pending		☒ COLLABORATIVE STATUS			
☒ TYPE OF PROPOSAL Research		A collaborative proposal from multiple organizations (PAPPG II.D.3.b)			
PI/PD DEPARTMENT Odum School of Ecology		PI/PD POSTAL ADDRESS 140 East Green St. Ecology Building Athens, GA 306022202 United States			
PI/PD FAX NUMBER 706-542-4819					
NAMES (TYPED)	High Degree	Yr of Degree	Telephone Number	Email Address	
PI/PD NAME Craig W Osenberg	PhD	1988	706-713-2051	osenberg@uga.edu	
CO-PI/PD					
CO-PI/PD					
CO-PI/PD					
CO-PI/PD					

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Yes ☐

No ☒

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- (1) has filed all Federal tax returns required during the three years preceding this certification;
- (2) has not been convicted of a criminal offense under the Internal Revenue Code of 1986; and
- (3) has not, more than 90 days prior to this certification, been notified of any unpaid Federal tax assessment for which the liability remains unsatisfied, unless the assessment is the subject of an installment agreement or offer in compromise that has been approved by the Internal Revenue Service and is not in default, or the assessment is the subject of a non-frivolous administrative or judicial proceeding.

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Certification Dual Use Research of Concern

By electronically signing the certification pages, the Authorized Organizational Representative is certifying that the organization will be or is in compliance with all aspects of the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern.

AUTHORIZED ORGANIZATIONAL REPRESENTATIVE		SIGNATURE		DATE
NAME Jennifer Jones		Electronic Signature		Aug 14 2018 12:05PM
TELEPHONE NUMBER 706-542-4575	EMAIL ADDRESS jllester@uga.edu		FAX NUMBER 706-542-5946	

COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./DUE DATE PD 98-1650 08/15/18		☐ Special Exception to Deadline Date Policy		FOR NSF USE ONLY NSF PROPOSAL NUMBER 1851268	
FOR CONSIDERATION BY NSF ORGANIZATION UNIT(S) (Indicate the most specific unit known, i.e. program, division, etc.) OCE - BIOLOGICAL OCEANOGRAPHY					
DATE RECEIVED	NUMBER OF COPIES	DIVISION ASSIGNED	FUND CODE	DUNS# (Data Universal Numbering System)	FILE LOCATION
08/15/2018	2	06040000 OCE	1650	053785812	08/16/2018 2:30pm S
EMPLOYER IDENTIFICATION NUMBER (EIN) OR TAXPAYER IDENTIFICATION NUMBER (TIN) 720423889		SHOW PREVIOUS AWARD NO. IF THIS IS ☐ A RENEWAL ☐ AN ACCOMPLISHMENT-BASED RENEWAL		IS THIS PROPOSAL BEING SUBMITTED TO ANOTHER FEDERAL AGENCY? YES ☐ NO ☒ IF YES, LIST ACRONYM(S)	
NAME OF ORGANIZATION TO WHICH AWARD SHOULD BE MADE Tulane University		ADDRESS OF AWARDEE ORGANIZATION, INCLUDING 9 DIGIT ZIP CODE Tulane University 6823 St Charles Avenue New Orleans, LA. 701185698			
AWARDEE ORGANIZATION CODE (IF KNOWN) 0020297000					
NAME OF PRIMARY PLACE OF PERF Tulane University		ADDRESS OF PRIMARY PLACE OF PERF, INCLUDING 9 DIGIT ZIP CODE Tulane University 6823 St Charles Ave New Orleans ,LA ,701185698 ,US.			
IS AWARDEE ORGANIZATION (Check All That Apply) ☐ SMALL BUSINESS ☐ MINORITY BUSINESS ☐ IF THIS IS A PRELIMINARY PROPOSAL THEN CHECK HERE ☐ FOR-PROFIT ORGANIZATION ☐ WOMAN-OWNED BUSINESS					
TITLE OF PROPOSED PROJECT Collaborative Research: Dynamic marine landscapes -- feedbacks and spatial patterns of corals and coral-associated fishes and invertebrates					
REQUESTED AMOUNT \$ 403,370	PROPOSED DURATION (1-60 MONTHS) 36 months	REQUESTED STARTING DATE 05/01/19	SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE		
THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW ☐ BEGINNING INVESTIGATOR ☐ HUMAN SUBJECTS Human Subjects Assurance Number _____ ☐ DISCLOSURE OF LOBBYING ACTIVITIES Exemption Subsection _____ or IRB App. Date _____ ☐ PROPRIETARY & PRIVILEGED INFORMATION ☒ INTERNATIONAL ACTIVITIES: COUNTRY/COUNTRIES INVOLVED ☐ HISTORIC PLACES FP ☒ VERTEBRATE ANIMALS IACUC App. Date <u>Pending</u> ☒ COLLABORATIVE STATUS PHS Animal Welfare Assurance Number _____ ☒ TYPE OF PROPOSAL Research A collaborative proposal from multiple organizations (PAPPG II.D.3.b)					
PI/PD DEPARTMENT Mathematics		PI/PD POSTAL ADDRESS 6823 St Charles Ave			
PI/PD FAX NUMBER 504-865-5063		New Orleans, LA 70118 United States			
NAMES (TYPED)	High Degree	Yr of Degree	Telephone Number	Email Address	
PI/PD NAME Scott A McKinley	PhD	2006	504-865-5727	smckinl3@tulane.edu	
CO-PI/PD					
CO-PI/PD					
CO-PI/PD					
CO-PI/PD					

CERTIFICATION PAGE

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Drug Free Work Place Certification

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Debarment and Suspension Certification

(If answer "yes", please provide explanation.)

Is the organization or its principals presently debarred, suspended, proposed for debarment, declared ineligible, or voluntarily excluded from covered transactions by any Federal department or agency?

Yes ☐

No ☒

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Certification for Contracts, Grants, Loans and Cooperative Agreements

The undersigned certifies, to the best of his or her knowledge and belief, that:

- (1) No Federal appropriated funds have been paid or will be paid, by or on behalf of the undersigned, to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with the awarding of any Federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.
- (2) If any funds other than Federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with this Federal contract, grant, loan, or cooperative agreement, the undersigned shall complete and submit Standard Form-LLL, "Disclosure of Lobbying Activities," in accordance with its instructions.
- (3) The undersigned shall require that the language of this certification be included in the award documents for all subawards at all tiers including subcontracts, subgrants, and contracts under grants, loans, and cooperative agreements and that all subrecipients shall certify and disclose accordingly.

This certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this certification is a prerequisite for making or entering into this transaction imposed by section 1352, Title 31, U.S. Code. Any person who fails to file the required certification shall be subject to a civil penalty of not less than \$10,000 and not more than \$100,000 for each such failure.

Certification Regarding Nondiscrimination

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is providing the Certification Regarding Nondiscrimination contained in Exhibit II-6 of the Proposal & Award Policies & Procedures Guide.

Certification Regarding Flood Hazard Insurance

Two sections of the National Flood Insurance Act of 1968 (42 USC §4012a and §4106) bar Federal agencies from giving financial assistance for acquisition or construction purposes in any area identified by the Federal Emergency Management Agency (FEMA) as having special flood hazards unless the:

- (1) community in which that area is located participates in the national flood insurance program; and
- (2) building (and any related equipment) is covered by adequate flood insurance.

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) or Individual Applicant located in FEMA-designated special flood hazard areas is certifying that adequate flood insurance has been or will be obtained in the following situations:

- (1) for NSF grants for the construction of a building or facility, regardless of the dollar amount of the grant; and
- (2) for other NSF grants when more than \$25,000 has been budgeted in the proposal for repair, alteration or improvement (construction) of a building or facility.

Certification Regarding Responsible Conduct of Research (RCR)

(This certification is not applicable to proposals for conferences, symposia, and workshops.)

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with the NSF Proposal & Award Policies & Procedures Guide, Chapter IX.B., the institution has a plan in place to provide appropriate training and oversight in the responsible and ethical conduct of research to undergraduates, graduate students and postdoctoral researchers who will be supported by NSF to conduct research. The AOR shall require that the language of this certification be included in any award documents for all subawards at all tiers.

CERTIFICATION PAGE - CONTINUED**Certification Regarding Organizational Support**

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that there is organizational support for the proposal as required by Section 526 of the America COMPETES Reauthorization Act of 2010. This support extends to the portion of the proposal developed to satisfy the Broader Impacts Review Criterion as well as the Intellectual Merit Review Criterion, and any additional review criteria specified in the solicitation. Organizational support will be made available, as described in the proposal, in order to address the broader impacts and intellectual merit activities to be undertaken.

Certification Regarding Federal Tax Obligations

When the proposal exceeds \$5,000,000, the Authorized Organizational Representative (or equivalent) is required to complete the following certification regarding Federal tax obligations. By electronically signing the Certification pages, the Authorized Organizational Representative is certifying that, to the best of their knowledge and belief, the proposing organization:

- (1) has filed all Federal tax returns required during the three years preceding this certification;
- (2) has not been convicted of a criminal offense under the Internal Revenue Code of 1986; and
- (3) has not, more than 90 days prior to this certification, been notified of any unpaid Federal tax assessment for which the liability remains unsatisfied, unless the assessment is the subject of an installment agreement or offer in compromise that has been approved by the Internal Revenue Service and is not in default, or the assessment is the subject of a non-frivolous administrative or judicial proceeding.

Certification Regarding Unpaid Federal Tax Liability

When the proposing organization is a corporation, the Authorized Organizational Representative (or equivalent) is required to complete the following certification regarding Federal Tax Liability:

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that the corporation has no unpaid Federal tax liability that has been assessed, for which all judicial and administrative remedies have been exhausted or lapsed, and that is not being paid in a timely manner pursuant to an agreement with the authority responsible for collecting the tax liability.

Certification Regarding Criminal Convictions

When the proposing organization is a corporation, the Authorized Organizational Representative (or equivalent) is required to complete the following certification regarding Criminal Convictions:

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that the corporation has not been convicted of a felony criminal violation under any Federal law within the 24 months preceding the date on which the certification is signed.

Certification Dual Use Research of Concern

By electronically signing the certification pages, the Authorized Organizational Representative is certifying that the organization will be or is in compliance with all aspects of the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern.

AUTHORIZED ORGANIZATIONAL REPRESENTATIVE		SIGNATURE		DATE
NAME Norey B Laug		Electronic Signature		Aug 15 2018 10:50AM
TELEPHONE NUMBER 504-865-5272	EMAIL ADDRESS norey@tulane.edu		FAX NUMBER 504-862-8674	

PROJECT SUMMARY

Overview:

Many nearshore marine ecosystems are defined by habitat-forming species (e.g., kelp, oysters, and coral) that provide shelter and food to other organisms. As a result, spatial variation in the amount, size and distribution of habitat-forming species plays a central role in determining patterns of density and diversity of associated fauna. While it is well-known that many occupants of biogenic habitat also affect the dynamics of the habitat itself (e.g., through consumption or fertilization), the role of bidirectional habitat-occupant interactions in driving the dynamics and spatial patterns of occupants and their habitat is poorly understood. The proposed research addresses these feedbacks in coral reef ecosystems. Specifically, we will study bidirectional interactions between corals in the genus *Pocillopora* (an important biogenic habitat) and the coral-associated fishes and invertebrates (CAFI) that depend on these corals for their persistence. The project uses field surveys, mesocosm and field experiments, and mathematical models to investigate the feedbacks in the coral-CAFI system by: (1) quantifying variation in the coral-CAFI spatial landscape; (2) testing the effects of habitat attributes (coral density, size, and proximity) on the settlement of CAFI and on the distribution of signals used by larvae to locate habitat; (3) testing the effects of habitat attributes on CAFI density (incorporating effects on settlement and post-settlement processes), on coral condition, growth, survival, and on the services that CAFI provide to the coral; and (4) extrapolating these shorter-term effects to understand long-term (e.g., multi-decadal) dynamics. This will be accomplished by building, parameterizing, evaluating, and applying a model that translates short-term experimental results into predictions of coral-CAFI dynamics on larger temporal and spatial scales. The model will be used to evaluate how coral ecosystems recover from disturbances, in light of the coral-CAFI feedbacks.

Intellectual Merit:

The mechanisms that underlie the effects of habitat configuration on the density of its occupants remain poorly understood. The proposed research tackles this problem by proposing and testing a mechanism (which we term "propagule redirection") that has the potential to explain diverse empirical relationships between habitat amount, patch size, and proximity on occupant density. Furthermore, the feedbacks that arise from the effects of occupants on their habitat are largely absent from past studies of landscape dynamics. Because variation in habitat attributes drives variation in occupant densities, which in turn likely drives the dynamics and distribution of the habitat itself, such feedbacks are a critical, albeit currently missing, component of landscape ecology. Further study of this connection will offer novel understanding regarding the creation of spatial patterns in habitat and associated fauna, and illuminate consequences of habitat loss and fragmentation that many systems defined by habitat-forming species are currently experiencing.

Broader Impacts:

This project will: (1) facilitate restoration effects; (2) enhance the scientific workforce; and (3) engage the public in scientific research and knowledge. For example, results of this project will provide insights about the restoration, conservation, and ecosystem-based management of coral reefs, a threatened and economically important ecosystem. Through a continuing collaboration with the NOAA Ecosystem Science Program, scientific outputs will be connected with opportunities to improve marine conservation and policy. Further, by coordinating with the Moorea Coral Reef LTER program, the project will benefit (and benefit from) ongoing studies of that ecosystem. This award facilitates NSF's Strategic Framework for Investment in Graduate Education by training several early career researchers including one postdoc, two PhD students, and several undergraduates. Students will be fully engaged in the research and will present products of this project at several venues, including national conferences. To include personnel from diverse backgrounds, PIs will coordinate with recruitment programs at UCSB, UGA, and Tulane (e.g., UCSB's Bridge to Doctorate Program). Outreach to the public will be conducted in French Polynesia (e.g., via the Atitia Center) and at the PI's home institutions, and will include a range of activities, including science fairs, ocean awareness programs, and community engagement projects designed to link arts and science.

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Table of Contents	1	_____
Project Description (Including Results from Prior NSF Support) (not to exceed 15 pages) (Exceed only if allowed by a specific program announcement/solicitation or if approved in advance by the appropriate NSF Assistant Director or designee)	15	_____
References Cited	7	_____
Biographical Sketches (Not to exceed 2 pages each)	4	_____
Budget (Plus up to 3 pages of budget justification)	6	_____
Current and Pending Support	2	_____
Facilities, Equipment and Other Resources	2	_____
Special Information/Supplementary Documents (Data Management Plan, Mentoring Plan and Other Supplementary Documents)	3	_____
Appendix (List below.) (Include only if allowed by a specific program announcement/ solicitation or if approved in advance by the appropriate NSF Assistant Director or designee)	_____	_____
Appendix Items:		

*Proposers may select any numbering mechanism for the proposal. The entire proposal however, must be paginated. Complete both columns only if the proposal is numbered consecutively.

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A. OVERVIEW AND OBJECTIVE

The need. Many marine systems are characterized by habitat-forming foundation species, which harbor a diversity of occupants, and whose dynamics are thought to drive resilience of entire ecosystems (Dayton 1972, Angelini et al. 2011). As a result, there is widespread concern over the ongoing loss and fragmentation of biogenic habitats such as seagrass beds, oyster reefs, kelp forests, and coral reefs (Graham 2004, Day et al. 2007, Beck et al. 2011, Byrnes et al. 2011, Angelini and Silliman 2012). Yet, without a more complete understanding of marine landscape ecology, we struggle to predict how the degradation or restoration of habitat will alter ecosystem dynamics, function, and resilience.

The gap. Most research in marine landscape ecology has focused on spatial patterns of occupant abundance and biodiversity; however, the causes and consequences of these patterns are rarely explored (Pittman et al. 2011). For example, many studies have shown that occupant *abundance* increases as the amount of habitat increases. What is less appreciated is that this pattern is often nonlinear (Fig 1A), which causes the *density* of occupants (per unit habitat) to decrease as the amount of habitat increases (Fig. 1B). However, we do not understand the mechanisms that cause this nonlinearity; nor do we understand the consequences of the variation in occupant density produced by the nonlinearity. An important but understudied consequence of variation in occupant density is that it may alter how occupants interact with their biogenic habitat. Because occupants can benefit biogenic habitat (e.g., by providing nutrients or protecting it from consumers) or harm biogenic habitat (e.g., by consuming or disturbing it), changes in occupant density can affect habitat growth and survival. Consequently, habitat-driven variation in occupant density should feed back to alter habitat dynamics and the spatial patterning of the habitat. In summary, we are limited in our understanding of *why* patterns in landscape ecology exist, *how* these patterns alter the population dynamics and spatial patterns of the occupants as well as their habitat, and *what* the implications of habitat degradation or restoration will be.

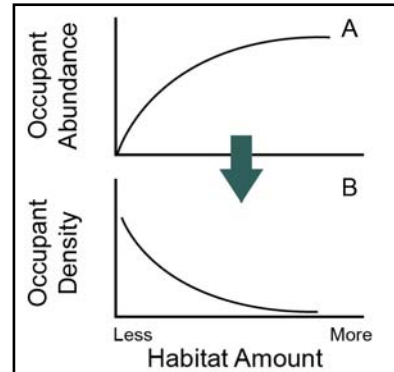


FIGURE 1. (A) The nonlinear relationship between occupant abundance and habitat amount, (B) produces a decreasing relationship between occupant density (no. per unit habitat) and habitat.

The objective. The central objective of this proposal is to examine the causes and consequences of the nonlinear relationship between occupant abundance and the amount of biogenic habitat. Specifically, we will: (i) *examine the habitat-based mechanisms that produce spatial variation in occupant density;* (ii) *quantify how habitat-driven occupant density feeds back to alter habitat growth and survival;* and (iii) *apply this knowledge to understand how bidirectional habitat-occupant interactions affect the long-term dynamics, create novel spatial patterns, and drive variation in how systems respond to and recover from disturbances.*

B. BACKGROUND

Above we focus on the general link between habitat amount and occupant density. Below, we focus on the landscape ecology of coral reef ecosystems. We broaden our discussion beyond the amount of habitat to consider three attributes of the coral habitat: the total **amount** of coral, the **size** of a coral colony, and the **proximity** of a colony to other colonies. We refer to these collectively as “habitat attributes” that together determine the habitat configuration. We also refer to occupants of coral colonies as **coral-associated fishes and invertebrates (CAFI)**, organisms that live within and interact closely with coral habitat. Throughout we use CAFI *abundance* to refer to the number of CAFI in a landscape and CAFI *density* to refer to the number of CAFI per unit of habitat.

Below we review previous research linking coral attributes to patterns of CAFI density (B-1). We then introduce our recently developed spatially explicit and mechanistic model that qualitatively reproduces empirical coral-CAFI patterns (B-2). Lastly, we detail how we will expand our study of marine landscapes to consider bidirectional interactions between corals and CAFI (B-3).

B-1. Coral attributes affects the density of CAFI. The amount (De'ath et al. 2012, Lamy et al. 2015), size, and proximity (Meesters et al. 2001, Karlson et al. 2007, Crabbe 2009) of coral varies widely, and these coral attributes are often strongly correlated to the density of CAFI (Jones 1997, Stier and Osenberg 2010).

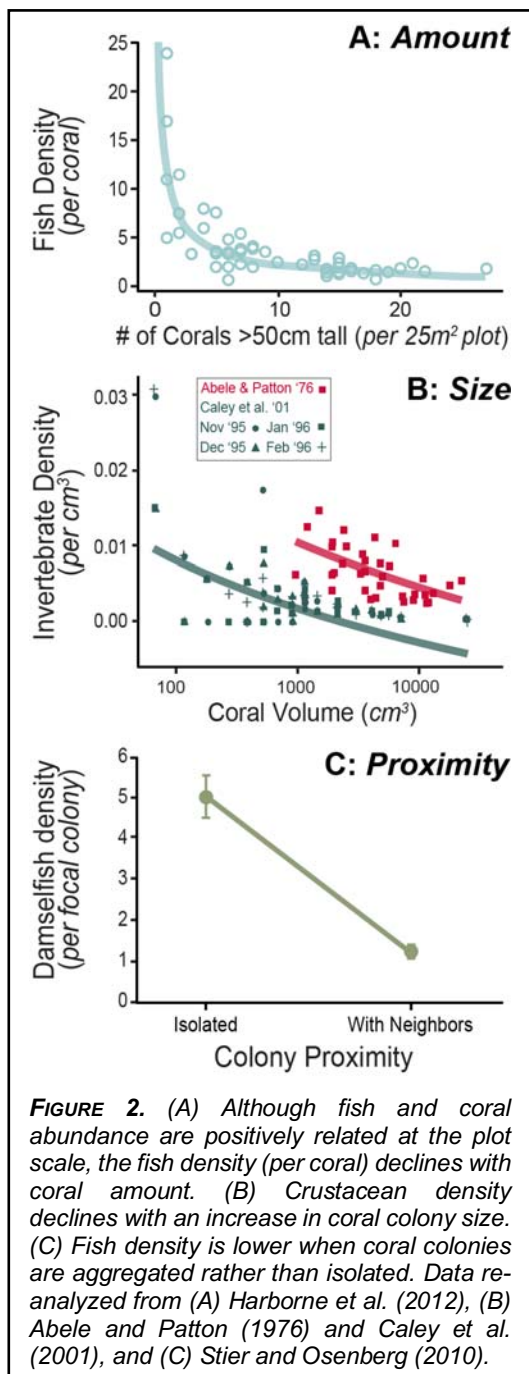
Habitat amount: Areas with more coral have greater abundances but lower densities of CAFI. The amount of coral habitat is an important driver of CAFI abundance (Holbrook et al. 2000). However, the amount of coral and the abundance of CAFI don't necessarily scale proportionately. For example, Harborne et al. (2012) showed that sites with more coral colonies had a greater abundance of fish; however, re-analysis shows that sites with more corals and more fish actually had lower **densities** of fish per unit habitat (Fig. 2A), suggesting that fish are "diluted" across available habitat.

Habitat size: Larger corals have greater abundances but lower densities of CAFI. Larger coral colonies typically support greater numbers (and species) of fishes and invertebrates (Abele and Patton 1976, Caley et al. 2001, Bonin et al. 2011); however, the size of coral and the abundance of CAFI don't necessarily scale proportionately. For example, in the South Pacific, CAFI density declines with increasing coral size (Fig. 2B). As a result, the way in which the total amount of coral is partitioned among corals of different sizes will greatly shape the CAFI assemblage.

Habitat proximity: Isolated corals have greater abundances and densities of CAFI. The proximity of a coral to other corals can further influence CAFI abundance, diversity, and composition. For example, we showed that isolated corals (i.e. without neighboring corals) had 2-5x greater settlement of CAFI than did corals with neighbors (Fig. 2C) (Stier and Osenberg 2010). In a related study, Jones (1997) found that recruitment of *Pomacentrus amboinensis* was over 2x greater to reefs and colonies that did not have an upstream neighbor. Collectively, these studies demonstrate that the proximity and the spatial arrangement of corals can drive spatial heterogeneity in CAFI settlement and density.

In summary, variation in coral amount, size, and proximity drives substantial variation in CAFI density. This motivates two additional questions:

- i) Is there a common mechanistic framework that can explain this variation in CAFI density?
- ii) What are the implications of this variation in CAFI density?



B-2. A spatially explicit and mechanistic model of coral effects on CAFI dynamics. To date, most studies of marine landscape ecology are empirical, and we know of no mechanistic framework that can reproduce the observed relationships between habitat attributes and occupant density (Fig. 2). We have developed a spatially explicit model of the coral-CAFI system that incorporates effects of coral attributes on CAFI settlement and the subsequent effects of CAFI settlement density on their post-settlement survival (Box 1).

Box 1. Our coral→CAFI model (based on Hamman et al. 2018) has several components:

(1) Larval CAFI settlement process (through propagule redirection, Fig. 3):

- (A) Propagules arrive in lunar pulses according to a spatial Poisson process (at concentration L).
- (B) Each coral, of radius R , emits a signal at a concentration $m(R)$ that diffuses away from the coral and degrades with time (degradation causes the process to deviate from pure diffusion), leading to a concentration approximated by $m(R)e^{-\rho d}$, where ρ is the decay rate (determined by degradation rate and diffusivity) and d is distance from the coral.
- (C) A larva successfully detects corals if the combined signal strength exceeds a threshold, θ .
- (D) The putative settler chooses a specific coral in proportion to its relative signal strength.
- (E) Larval survival during transit to the coral is $e^{-\mu d}$, where μ is the per-distance mortality rate.

(2) CAFI post-settlement process:

Post-settlement survival is described by a Beverton-Holt recruitment function, in which the per capita mortality rate equals $-\alpha - \beta N$, where α and β are the density-independent and dependent mortality rates, and N is the occupant density (accumulated from prior settlement pulses).

Steps 1-2 are repeated for multiple settlement pulses (larval delivery is episodic [discrete]; while mortality occurs continuously). All processes are stochastic.

Hamman et al (2018) solved portions of the model analytically (through deterministic approximation) and simulated the fully stochastic version in static landscapes that differed in the spatial arrangement of corals (and in a landscape based upon the north shore of Moorea). Resulting spatial variation in CAFI can be substantial, reflecting the underlying coral landscape (Fig. 3,4).

Our framework is motivated by empirical data on the effects of habitat amount and proximity on settlement of CAFI (Fig. 2C, Stier and Osenberg 2010). Specifically, larvae settling to a coral landscape experience “**propagule redirection**” in which neighboring corals “compete” for settling larvae (Carr and Hixon 1997, Osenberg et al. 2002a, Stier and Osenberg 2010). Larval CAFI enter the landscape and then locate suitable settlement sites using chemical cues emitted from corals (e.g., Lecchini et al. 2005, Foretich et al. 2017). An isolated coral draws in larvae from a large area, but when there are many corals in the local neighborhood, the pool of competent larvae gets distributed amongst the available corals (Fig. 3, 4). Propagule redirection thus establishes settlement patterns, which are subsequently modified by post-settlement mortality (Hixon and Webster 2002, Osenberg et al. 2002b; see Hamman et al. 2018). Parameterization of our model using empirical estimates of propagule redirection (Stier and Osenberg 2010), density dependence (Osenberg et al. 2002b), and habitat quality (Shima and Osenberg 2003) predicts that these patterns are not simply transient phenomena caused by habitat-driven settlement, but instead drive persistent spatial variation in the density of

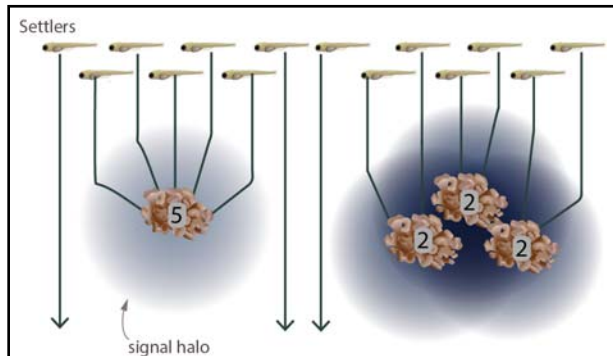


FIGURE 3. Propagule redirection. Signals emitted by corals attract settlers, but the pool of settlers drawn to the signal gets divided among the available corals. As a result, a group of corals (right) receive more settlers, but achieves a lower density compared to an isolated coral (left). This creates spatial variation in CAFI density even when corals are identical. Signal halo is drawn illustratively and without regard to modification by flow (which is indicated by the direction of larval movement).

adult CAFI (Hamman et al. 2018). For example, our model predicts that the equilibrium CAFI density in an isolated coral could be >10-fold greater than in a coral with many neighbors, despite the potentially homogenizing effects of density-dependence.

Our model reproduces the qualitative effects of coral attributes: CAFI density declines with increasing number of coral colonies at a site (Fig. 2A), increases in colony size (Fig. 2B), and increasing proximity to neighboring corals (Fig. 2C, 4).

B-3. The next steps – feedbacks between CAFI density and coral growth, survival, and condition.

Currently, the model (Hamman et al. 2018) focuses on the effects of corals on CAFI, while ignoring possible feedbacks of these resident CAFI on corals. Yet CAFI can have large effects on the growth, survival, and condition of their coral host. For example, crustaceans promote coral survival by defending the corals from predatory sea stars and removing sediment and other debris from host colonies (Glynn 1983, Stewart et al. 2006, Stier et al. 2010, 2012, McKeon et al. 2012). Fish increase coral growth by subsidizing nitrogen via excretion (Meyer and Schultz 1985, Holbrook et al. 2008). Conversely, other CAFI may harm corals (Stier et al. 2010, Shima et al. 2010, Enochs and Glynn 2017). For example, corallivorous gastropods such as *Drupella* spp. consume coral tissue, reduce coral growth (Boucher 1986, Turner 1997, Shafir et al. 2008, Hamman 2018), and facilitate disease transmission (Nicolet et al. 2013). The intensity of these beneficial and deleterious effects increases with CAFI density. Based upon the effects of coral attributes on CAFI density (e.g., Fig. 2, 4), we expect that CAFI will have stronger (i.e., more deleterious or more beneficial) effects when there is less coral, corals are small, and corals are isolated. However, we are unaware of comprehensive empirical or theoretical studies examining how bidirectional habitat-occupant interactions drive the dynamics of landscapes, either in coral reefs or other systems.

C. HYPOTHESIS & AIMS

Hypothesis: Variation in coral amount, size, and proximity drives variation in CAFI density, and this variation in CAFI density produces variation in coral growth and survival. This bidirectional interaction between corals and CAFI will fundamentally alter the dynamics of marine landscapes.

To evaluate this hypothesis and understand the long-term consequences of the bidirectional coral-CAFI interactions we will accomplish the following 4 specific aims:

- ◆ **Aim 1 (Landscape characterization):** Characterize variation in coral attributes (amount, distribution, size, and proximity) and CAFI (composition, abundance, and their relationship to the coral landscape).
- ◆ **Aim 2 (Short-term, colonization experiments):** Experimentally quantify how coral attributes (number of colonies, size, and proximity) produce variation in settlement of CAFI, and evaluate how these effects are modified by water flow and mediated by the distribution of chemical signals.
- ◆ **Aim 3 (Long-term experiment):** Quantify the consequences of habitat-driven shifts in CAFI settlement on long-term CAFI density and coral performance (growth, survival, and condition).
- ◆ **Aim 4 (Model):** Build, parameterize, and evaluate a spatial model of bidirectional coral-CAFI interactions; use the model to explore long-term dynamics and spatial patterning of corals and CAFI; and evaluate how coral-CAFI feedbacks affect coral recovery from disturbance.

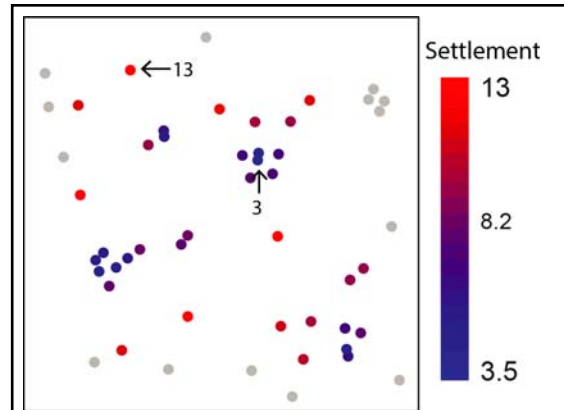
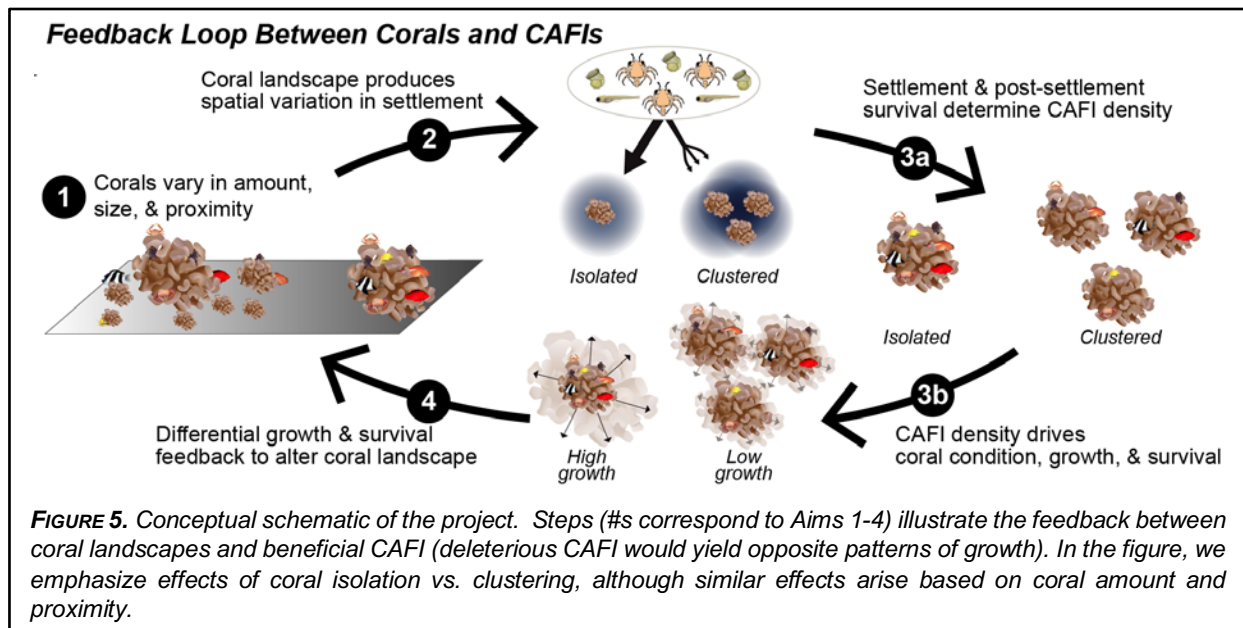


FIGURE 4. Variation in coral proximity drives variation in settlement due to propagule redirection. Dots represent corals (identical except for spatial positioning) with colors indicating the intensity of CAFI settlement. Gray circles are neighboring corals that affect settlement but are not part of focal landscape (and were omitted due to edge effects). Arrows indicate corals that had the minimum and maximum CAFI density in the landscape. Patterns persist after post-settlement processes based on empirical estimates of density-dependent mortality. Results based on Hamman et al (2018).



D. RESEARCH PLAN

Study Site. The proposed studies will take place in Moorea, French Polynesia, benefiting from our extensive knowledge of the system, expertise provided by the Moorea Coral Reef LTER, and our past success working in the region. Following an outbreak of predatory sea stars and a hurricane, *Pocillopora* spp.¹, which host a high diversity of fishes and invertebrates (Stier and Leray 2014), have increased over 1000-fold and are the primary branching coral in the lagoon (Tsounis and Edmunds 2016), providing an abundant habitat for our studies. These features make the *Pocillopora*-CAFI system in Moorea ideal for studying how the feedbacks between corals and CAFI influence the landscape ecology of coral reefs.

Aim 1 (Landscape Characterization): Quantify the spatial landscape of coral-CAFI interactions

The objectives of Aim 1 are to (i) document how coral landscapes vary in amount, size, and colony proximity and (ii) link these habitat attributes to the community structure of CAFI and the condition of the corals. We will use underwater visual censuses to quantify the spatial co-variation in habitat attributes (*Pocillopora* amount, size, and proximity), CAFI community characteristics (density, diversity, and distribution), and coral condition. Our working hypotheses, motivated by empirical work (e.g., Fig. 2 & 3; B-1) and the model of Hamman et al. (2018) (B-2), are that: (i) larger corals will have greater CAFI abundances but lower CAFI densities; (ii) more isolated corals will have greater CAFI abundances and densities than corals with many neighbors; (iii) within a site, plots with greater coral cover will have greater CAFI abundances but lower CAFI densities; and (iv) corals with greater densities of beneficial CAFI (e.g., fishes and crabs) should be in better condition than corals with few beneficial CAFI, or than corals with greater densities of deleterious CAFI (e.g., *Drupella*) (B-3). Our approach involves quantifying: coral landscapes (Activity 1-1), densities and composition of CAFI (Activity 1-2), coral condition and morphology (Activity 1-3); and the spatial covariance among these variables. Upon completion of this field survey, we expect to have a clear understanding of how variation in CAFI density and community composition is correlated with coral attributes (amount, size and proximity) and environmental features that vary among sites (e.g., current speed and settlement intensity), and how these factors combine to influence coral condition.

¹ Several species of *Pocillopora* occur in Moorea, although the taxonomy is challenging due to cryptic species and high plasticity (Forsman et al. 2013, Edmunds et al. 2016, Johnston et al. 2017). We will sample all specimens that occur in our surveys, classify by morphotypes and inspect calyx structure. Our experiments will target a single morphotype (arguably *P. verrucosa*: Johnston et al. 2017).

Activity 1-1. Mapping the spatial landscape of *Pocillopora*.

We will conduct surveys on SCUBA to characterize the amount, sizes and proximity of *Pocillopora* coral colonies in the backreef of Moorea. At each of 9 sites that define a gradient of CAFI settlement intensity and vary in current intensity (Schmitt and Holbrook 1999), we will measure the circumference & height of all *Pocillopora* colonies within two 10x10m plots, which will be recorded as a map. The amount and scale of spatial autocorrelation within a study plot will be quantified using Moran's I and Geary's c (Fortin et al. 2002). Size distributions will be compared among plots using Kolmogorov-Smirnov tests (Darling 1957).

Activity 1-2. Overlaying the distribution of CAFI on the coral landscape.

Within each of the surveyed plots, divers will visually identify and count fishes, crustaceans, and gastropods living within the *Pocillopora* colonies that occupy the central 5x5m² portion of the plot (the surrounding 5m is needed as a buffer to ensure that we quantify neighborhoods of each focal coral). We will sample up to 20 of these focal corals by placing a bag around each coral, removing it from the substrate, and transferring it (and its CAFI) to a 5 gallon bucket (Stier and Leray 2014). Our subsample of colonies will encompass a range of coral sizes and degree of isolation from other corals. On the boat, we will remove all CAFI using 0.02% eugenol (Leray et al. 2012). In the lab, CAFI will be identified, counted, and weighed.

To test for effects of coral configuration, we will use mixed effects multiple linear regression with coral size and proximity as fixed effects, site as a random effect, and CAFI density as the response variable. Proximity will be estimated using a metric that sums across the landscape, downweighting neighbors by distance and upweighting by size, as suggested by (Hamman et al. 2018): see also Silander and Pacala (1985), Coates et al. (2009). We also will examine the use of covariates such as wave power and current velocity available from MCR-LTER monitoring. To examine differences in species composition we will use Principal Coordinate Analysis (PCoA) to characterize CAFI community composition, and then use Mantel tests and multiple regression to quantify the influence of coral habitat (amount, size, proximity) and environmental traits (e.g., current speed).

ACTIVITY 1-3. Quantifying coral traits in relationship to the coral-CAFI landscape.

Two fragments will be removed from each coral just after collection from study plots (but before removal of CAFI) and airbrushed to remove all tissue; the homogenized tissue slurry will be subdivided for analysis of lipid, protein and carbohydrate content, tissue biomass, chlorophyll content, and *Symbiodinium* density. Surface area of fragments will be assessed via the wax dipping technique (Veal et al. 2010). *Symbiodinium* density will be quantified from cell counts on a hemocytometer and normalized to surface area and ash free dry mass (AFDM). Chlorophyll will be extracted in 100% acetone in the dark for 24h. Chlorophyll *a* and *c₂* content will be measured on a spectrophotometer at wavelengths of 630, 663, and 750 nm (blank), and normalized to surface area, AFDM, and *Symbiodinium* cell density (Jeffrey and Humphrey 1975, Putnam and Edmunds 2011). Lipids will be extracted in a 2:1 chloroform-methanol solution and washed in 0.88% KCl, 100% chloroform, and 0.88% KCL, dried at 60°C, weighed, and normalized to AFDM (Harland et al. 1992). Colorimetric analysis of coral carbohydrate content will be conducted using the phenol-sulfuric acid method with D-glucose as a standard (Dubois et al. 1956). Total protein content will be determined using bicinchoninic acid (BCA) as a detection reagent, and bovine serum albumin as reference after adding 0.1M NaOH, heat treated to 50°C for 5 hours, neutralized with 0.1M HCl, and normalized to surface area and AFDM. Statistical tests will be conducted as in Activity 1-2, except that we will analyze the coral traits as functions of CAFI in addition to coral attributes.

Aim 2 (Field Experiments): A mechanistic framework linking habitat attributes and currents with settlement of CAFI

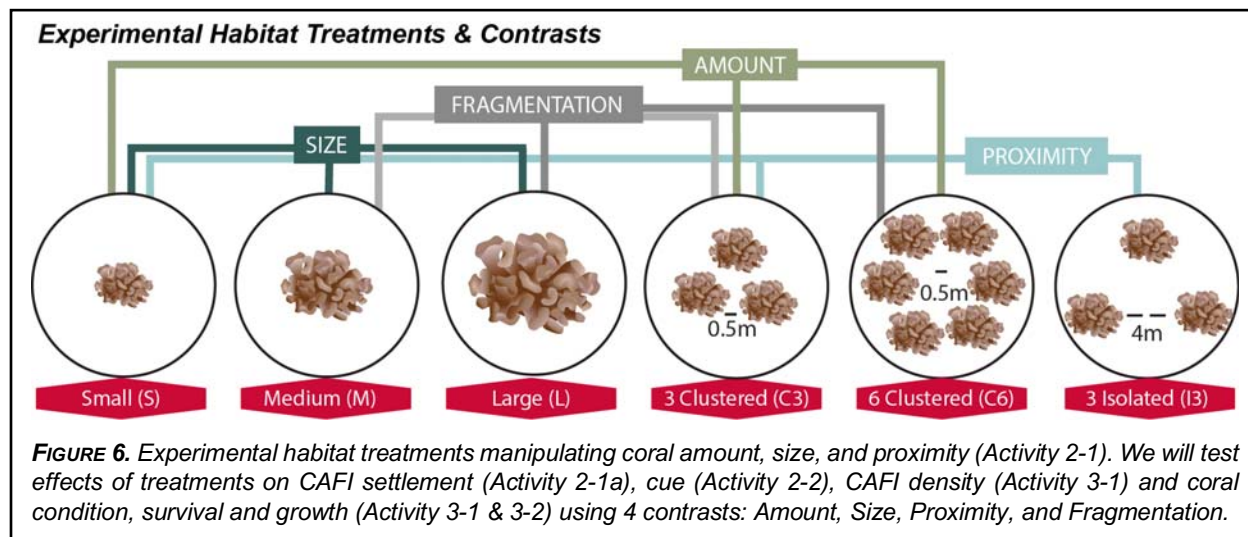
The objective of Aim 2 is to quantify how habitat attributes and current speed alter the strength of propagule redirection to determine CAFI settlement patterns. Our field and modeling studies suggest that propagule redirection is the primary *cause* of occupants responding disproportionately to changes in habitat (Fig. 2). However, our understanding of this mechanism is incomplete: e.g., we do not know the spatial scale over which redirection operates nor how this is modified by the physical environment. The new data will provide the foundation for a mechanistic model of settlement (in Aim 4), which in combination with post-settlement

processes determines CAFI spatial distributions and the effects of CAFI on coral condition, growth and survival.

Our working hypothesis is that the settlement of CAFI (expressed per unit habitat) will decline with increasing amount, size, and proximity of coral. Our subsidiary hypothesis is that currents will reduce the strength of these effects by lowering the amount of overlap among coral signals. Our approach consists of three replicated, short-term, field experiments (Activity 2-1) to quantify how coral attributes (amount, size, proximity) and currents alter the strength of redirection in driving CAFI settlement. Each experiment will be conducted at the 9 sites surveyed in Activity 2-1 and will span two lunar cycles. In Activity 2-2, we will quantify the "signal halo" that characterizes settler-coral interaction – how far the signal extends around the coral. Later, in Aim 3, we explore the longer-term consequences of these settlement patterns.

ACTIVITY 2-1. Field tests of the effects of habitat attributes on CAFI settlement.

2-1a. Coral amount, size, and proximity. We will create six treatments with varied coral attributes (Fig. 5): (S) 1 small coral (30cm diameter); (M) 1 medium coral (43cm diameter – 3x the volume of S); (L) 1 large coral (54 cm diameter – 6x the volume of S); (C3) 3 clustered small corals (0.5m apart; same total volume as M); (C6) 6 clustered small corals (0.5m apart; same total volume as L); and (I3) 3 isolated small corals (4m apart). Treatments will be replicated unequally to ensure that the single corals are sampled sufficiently: 6 replicates of S, 3 replicates of M and L, and 2 replicates of C3, C6, and I3 per site. Standardized Habitat Units (SHUs) will be constructed using PVC collars and three bolts (i.e. a 'tree stand': Stier et al. 2010) to facilitate buoyant weight measurements at the conclusion of the experiment. To reduce the collection of corals from reef habitat, SHUs will be built with colonies collected during Activity 1-2. Each replicate will be at least 15m from natural hard substrate and other replicates. We will visit each of the SHUs daily, and use hand nets and anesthetic to collect fishes and invertebrates that settled the night before (Stier and Osenberg 2010). Collected settlers will be preserved, identified, counted, and measured in the lab. We will conduct four contrasts to test the effect of habitat attributes on settlement (Fig. 5): (1) size (S-M-L), (2) amount (S-C3-C6), (3) fragmentation (M-C3; L-C6), and (4) proximity (C3-I3-S: i.e., 0.5 vs. 4 vs. 15m).



2-1b. Coral proximity. To better define the spatial scale over which redirection operates, we will conduct a gradient experiment focused on coral proximity. At each of the 9 sites, we will have 7 treatments, each with 3 SHUs (each with a 30cm diameter coral -- to facilitate comparison with C3 in 2-1a) placed at different distances from one another: (a) 0.5m (identical to C3), (b) 1m, (c) 2m, (d) 4m (identical to I3), (e) 6m, (f) 10m, and (g) 15m (identical to S). We will measure relative current speed at each site over the duration of the experiment with clod cards (calibrated to the MCR-LTER monitoring stations). The data will be fit using our propagule redirection model, and used to estimate the shape of the relationship between settlement and proximity, the distance at which settlement becomes independent of neighbors, and the effect of current

speed on these relationships. We expect that among-site variation in currents will affect the distance over which corals compete for settlers (e.g., through effects on the Peclet number: see Activity 2-4).

2-1c. Larval depletion and water flow. Propagule redirection arises because competing signals from habitat patches give rise to a redistribution of settlers among available habitat (Stier and Osenberg 2010, Hamman et al. 2018). Settlement also depletes the pool of competent larvae from the water column leading to *recruitment shadows*, which are evident in comparing upstream vs. downstream sites (Gaines et al. 1985, Jones 1997). To date, we have not incorporated the effect of flow in our model of propagule redirection, but evidence suggests flow is likely important (Jones 1997, Fig. 6). We will conduct a field experiment and quantify settlement to four 8x8 grids of SHUs, each with 2m between corals. We predict variation in settlement moving upstream to downstream will reflect effects of depletion (i.e., shadows), while propagule redirection alone will create the cross-current gradient (with the greatest settlement to the edge corals and reduced settlement to the central corals, which have more neighbors). A control treatment will have a single cross-current row of 8 corals (i.e., without the potential for recruitment shadows). This experiment is facilitated by the predominantly unidirectional currents in the Moorea lagoon (Hench et al. 2008). Data will be used to parameterize and evaluate the incorporation of flow into the model developed in Aim 4.

ACTIVITY 2-2. Quantify spatial patterns in signal strength as functions of coral amount, size, and proximity.

Diverse sensory systems are used by fishes and invertebrates to locate settlement habitat (e.g., Lecchini et al. 2005), and are likely responsible for many of the effects of habitat on settlement that our project will document. Because most CAFI settle at night to specific coral species, chemical cues play a particularly important role, especially at smaller spatial scales (Holbrook and Schmitt 1997, Dixon et al. 2014, Foretich et al. 2017). Although considerable work has been done to document the existence of cues (e.g., Lecchini et al. 2005), the spatial distribution of signals has received little to no attention. Parameters in our model define a "signal halo", which will determine how different coral spatial configurations will affect the distribution of settlers. To better understand mechanisms that create settlement patterns (Activity 2-1) we will measure the spatial distribution of dimethylsulfide (DMS) around focal corals. Although DMS is just one of a suite of chemicals that may serve as a settlement cue, it is one of the few compounds demonstrated to attract multiple taxa of larval fish on coral reefs (Foretich et al. 2017), and therefore is a useful target for our purposes; its distribution should correlate with those of other compounds that serve similar roles.

We will measure DMS by collecting water samples at points distributed around each coral treatment in Activities 2-1a,b,c (precise numbers and locations will be determined in pilot studies), and analyzing samples at the Gump lab immediately after collection following the methods of Nagahata et al. (2013). To quantify temporal consistency, we will replicate these measurements during the 2nd, 4th, and 7th week of each experiment. These samples will be accompanied by synoptic measurements of current speeds taken with handheld current meters (at 0.5 and 1m above the sediments adjacent to the SHUs) to improve estimation of signal propagation under varying physical conditions -- we expect that current speed will affect signal distribution via changes in advection. Measurements of DMS will be invaluable in parameterizing and evaluating our model of the signal landscape (Aim 4).

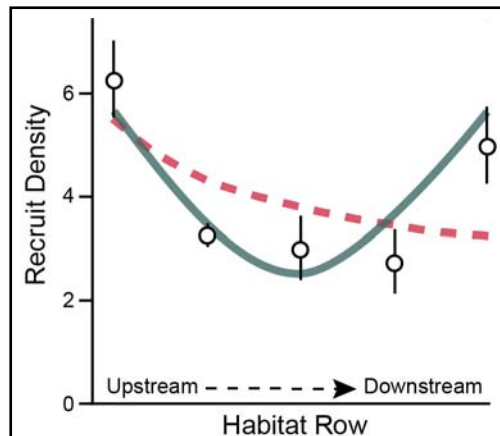


FIGURE 7. Recruit density upstream and downstream to corals parallel to prevailing currents (Jones 1997). **Solid line** is the prediction based upon our propagule redirection model without flow (Hamman et al. 2018); **dashed line** is the prediction based only on recruitment shadows (i.e., larval depletion: Jones 1997, Osenberg et al. 2002a). Data are best predicted from our redirection model, although the residuals suggest a need to incorporate flow.

Aim 3 (Field Experiment): Determine the longer-term consequences of settlement patterns

Aim 1 quantifies the coral landscape and establishes correlations between coral landscape attributes and patterns of CAFI densities and coral traits. Aim 2 explores the causative role of habitat amount, size, and proximity in driving CAFI settlement, and investigates the role of settlement cues in producing these patterns. The objective of Aim 3 is to quantify the longer-term *consequences* of the effects of coral attributes on CAFI settlement patterns. In particular we ask: (1) how does landscape-driven variation in CAFI settlement translate to long-term patterns of CAFI density; and (2) how does this variation in CAFI density correlate with coral performance and resilience to environmental stressors?

Our working hypothesis is that CAFI densities will resemble those created by propagule redirection but dampened by post-settlement processes: CAFI density will be lower in larger corals, in treatments with more coral, and in corals near other corals. Deviations between the settlement patterns observed in Aim 2 and occupancy patterns observed in Aim 3 will be attributed to post-settlement processes. Our subsidiary hypothesis is that CAFI density will be more homogeneous at sites with greater larval supply (as assessed in Activity 2-1) because stronger post-settlement density-dependence will reduce the spatial variation in settlement induced by the habitat (Hamman et al. 2018). These patterns in CAFI density and composition should be reflected in coral performance: e.g., if most CAFI are beneficial, then corals with more CAFI should exhibit greater growth, survival and condition. Our approach includes a year-long field experiment, whose design mirrors the experiment focused on settlement (Activity 2-1), to quantify the effects of habitat landscape on the density and composition of CAFI and on coral condition, growth, and survival (Activity 3-1). At the completion of the experiment, we will bring a subset of the corals and their CAFI into the lab to quantify the capacity of CAFI to protect the coral colony from predators and sedimentation (Activity 3-2).

ACTIVITY 3-1. Long-term spatial patterns in CAFI density and coral performance.

We will conduct a replicated field experiment to quantify the effects of habitat attributes as CAFI settle over multiple lunar cycles, incur post-settlement processes, and potentially affect coral condition, growth, and survival. The experiment will be conducted at the same 9 sites as in Activity 1-1 and 2-1, using the same 6 treatments as in Activity 2-1 (Fig. 5) to facilitate comparison of patterns evident immediately following settlement to those that are expressed after the effects of density-dependence accrue over multiple cohorts.

Prior to the start of the experiment, we will remove CAFI from each SHU, measure their initial volumes, and buoyantly weigh them to estimate skeletal mass (Davies 1989). The CAFI community will be visually censused by SCUBA divers on a weekly basis for the first month, on a monthly basis for the next three months, and then as permitted. During these censuses we will distinguish recent recruits to re-assess patterns observed in 2-1 and examine whether those patterns change temporally. Corals that die will be replaced (but tagged as replacements) to maintain treatments. After one year, we will conduct a final visual census, re-measure coral volume (to measure volumetric growth), and take a tissue sample (to measure coral condition: see methods in Activity 1-3). Half of the corals in each treatment will be collected (as in Activity 1-2), their CAFI removed using eugenol, and buoyantly weighed (to estimate skeletal growth). All CAFI will be identified, counted, and weighed. The remaining corals will be transferred to the Gump lab to assess performance (Activity 3-2: see below).

Effects on coral traits could result through two mechanisms: (1) indirect effects of coral treatments mediated through CAFI (our working hypothesis), or (2) direct effects of coral-coral interactions (e.g., due to increased competition for food at high local density). To evaluate direct coral-coral effects, we will assess changes in skeletal mass (using the buoyant mass technique) and tissue traits (see above) at the conclusion of the 60 day experiments in Activity 2-1. Because CAFI are removed daily in Activity 2-1, our working hypothesis is that there will be no among-treatment variation in coral growth or condition. If we observe treatment effects (supporting the coral-coral hypothesis) then we will modify Activity 3-1 based on those results (although chronic removal of CAFI over an entire year is not feasible). Fortunately, we think it is unlikely that coral-coral interactions will arise at scales $\geq 0.5\text{m}$.

ACTIVITY 3-2. Effects of CAFI on sediment removal and anti-predator defense.

Using the remaining corals from Activity 3-1, we will test our hypothesis that corals with higher density and diversity of beneficial CAFI will receive greater protective services than colonies housing lower density and

diversity of beneficial occupants (or those with deleterious CAFI). Corals and their CAFI communities will be maintained in individual 40L aquaria with flow-through seawater and evaluated using a sediment removal test and a corallivory test.

Approach. We will collect sediments from the lagoon, pass sediment through soil sieves to remove the >0.1 mm fraction (leaving a terrigenous silt/clay mix), then sterilize and dry the sediments. All corals will be exposed to sedimentation rates of $100 \text{ mg m}^{-2} \text{ day}^{-1}$, which is the upper range naturally observed in Moorea (Padilla-Gamiño et al. 2012, see also Zill et al. 2017). This test will allow us to quantify the capacity of CAFI to remove sediment, following the methods of Stier et al. (2012). After completing the sedimentation test, the anti-predator effectiveness of the CAFI assemblages will be assessed against a corallivorous sea star (*Culcita novaeguineae*) using the protocol developed by McKeon et al. (2012). After both tests, we will remove the CAFI using eugenol (Stier and Leray 2014) and enumerate their density, diversity, and composition.

Data Analysis. We will use parametric, nonparametric and multivariate analyses to evaluate effects of habitat treatments on CAFI density and composition, coral condition, growth and survival, and CAFI protective services. For example, comparative analyses across treatments will include generalized linear mixed models with CAFI density and coral condition as response variables, and PCoA and PERMANOVA to look at CAFI community composition. We will use multivariate tests to compare rates of sediment removal and predation across treatments. Additionally, stress test response variables will be regressed against CAFI density, diversity, and a PCoA axis of composition.

Aim 4 (Modeling): Integration across scales -- the long-term consequences of coral-CAFI interactions for coral and CAFI population dynamics, spatial patterning, and coral recovery

The *objective* of Aim 4 is to develop, evaluate, and apply a model of the feedbacks between corals and CAFI. We will build on our initial theoretical work (Hamman et al. 2018), which focused on the unidirectional effect of the coral landscape on CAFI distributions and dynamics, specifically by incorporating the other half of the feedback loop: the effects that CAFI have on corals. We seek to understand how these coral-CAFI feedbacks give rise to a wide array of possible spatial patterns and recovery dynamics in coral reef landscapes.

To illustrate a possible consequence of habitat-CAFI interactions, consider a simple thought experiment. First, recall that CAFI density tends to be greater when corals are smaller (Fig. 2B). If CAFI enhance coral growth (e.g. Holbrook et al. 2008), then larger coral colonies will benefit less from CAFI, and smaller colonies will benefit more, because CAFI density is lower on the large colonies. Thus, one *working hypothesis* is that *habitat-driven variation in the density of beneficial CAFI should homogenize the coral colony size distribution* (i.e., lead to lower variance in colony size compared to the situation without CAFI). On the other hand, if a particular CAFI species reduces coral growth, this should increase the variance in the coral colony size distribution.

Another thought experiment illustrates the potential effects of habitat-CAFI interactions on spatial patterns of coral colonies. CAFI density on an isolated coral can be much greater than the density on a coral with many neighbors (Fig. 2C, 4). When CAFI are beneficial, this will encourage growth and survival of isolated corals, while clustered corals will struggle. This mechanism gives rise to another *working hypothesis* that *beneficial interactions will lead to overdispersed coral landscapes*. By contrast, *deleterious interactions will lead to clustered coral landscapes*. This shows the pivotal role that propagule redirection (sensing and decision-making by larval settlers) might have on large-scale spatial patterns of habitat.

These thought experiments are just two of many intriguing relationships. With a well-constructed and flexible modeling framework – extending what we have already developed – we can (1) study the short-term effects that habitat amount, size, and proximity can have on the density of occupants, and ask what parameter combinations give rise to habitat-occupant patterns observed in nature; (2) use parameters inferred from empirical work in Aims 2 and 3 to determine how variation in coral amount, size, and proximity interact with the beneficial and deleterious effects of these CAFI to drive short-term patterns of coral growth; (3) describe long-term equilibrium dynamics and the emergent spatial distribution of coral and CAFI, contrasting patterns that result from beneficial vs. deleterious CAFI; and (4) simulate environmental

disturbances of different magnitude and type to examine how various coral-CAFI feedback scenarios result in different rates of, and spatial patterns in, recovery.

Our *approach* has three core components: (1) **model development** (including innovations needed to capture both the effect of flow on larval CAFI settlement and the effect of CAFI density on coral growth and survival), and **theoretical analysis** (featuring analysis of both stochastic models and deterministic approximations) (Activity 4-1); (2) Bayesian-style **parameter estimation** with qualitative **model checking**, using data from Aims 2 and 3 with short-time, small-scale versions of our model (Activity 4-2); and (3) **extrapolation** to questions that cannot be assessed in empirical studies lasting <3 years: e.g., to examine the **large-scale consequences of coral-CAFI feedbacks** for recovery dynamics following a system-wide disturbance.

ACTIVITY 4-1: Development of mathematical model and theoretical analysis

Because CAFI abundances per coral are relatively small (e.g., often <10), and the data collection techniques are intrinsically noisy, a *stochastic* modeling approach (including Markov chains and agent-based simulations) is necessary to emulate what is observed in the field. Beyond producing authentic-looking population counts and spatial distributions, stochastic models are essential for producing likelihood functions (or approximations thereof) for use in the Bayesian inference techniques described in Activity 4-2. We will partner these stochastic methods with more traditional *deterministic* techniques (differential equations) to facilitate analysis. For example, we will implement age- and/or size-structured models based on the framework presented in Box 2, with all stochastic effects replaced by appropriately averaged quantities. We will use that deterministic representation to obtain explicit solutions (when feasible) and identify different regions of parameter space that give rise to qualitatively distinct behaviors: e.g., in one region of parameter space, beneficial coral-CAFI feedbacks might lead to coral size distributions with low variance, while in another part of parameter space CAFI might lead to size distributions with high variance. With qualitatively distinct parameter regimes identified by the deterministic model, we will choose several representative parameter combinations from each regime and use the stochastic models to simulate coral-CAFI landscapes.

Box 2. Our new model includes both coral-CAFI interactions and propagule redirection. Each month:

- (1) Larval corals settle according to a spatial Poisson process (at concentration L_c).
- (2) Larval CAFI settle according to the propagule redirection process described in Step 1 of Box 1 with an extension that allows for flow in the environment (Activity 4-1, Innovation 2).
- (3) Settled CAFI undergo post-settlement mortality (Step 2 in Box 1).
- (4) Corals grow and survive at rates that depend on coral size and CAFI density (Activity 4-1, Innovation 1).

Steps 1-4 are repeated, and all processes are stochastic. Note that coral spatial configuration is an *emergent* property of the model. The challenge is to characterize how these patterns are driven by the beneficial/deleterious nature of coral-CAFI interactions.

We will develop several model innovations to extend our earlier efforts (Hamman et al. 2018). We highlight two innovations that seem to be particularly important.

Model Innovation 1: Coral-CAFI feedbacks. Our existing model assumes that the habitat is static and looks only at effects of habitat on the distribution and abundance of CAFI (Box 1) with parameter values inferred from our studies of density-dependence (Schmitt et al. 1999, Osenberg et al. 2002b) and habitat quality (Shima and Osenberg 2003, Shima et al. 2008). But, existing data demonstrate that CAFI also affect coral growth and survival (Stier et al. 2010, 2012, McKeon et al. 2012, Hamman 2018). We will contrast the effects of CAFI that increase or decrease coral skeletal growth and survival (e.g., fishes that excrete nutrients that promote coral growth vs. gastropods that consume coral tissue). We will address effects of specific combinations of species in future research. We begin by assuming that corals grow at some baseline size-dependent rate without symbionts, $G_0(R)$. For beneficial CAFI, the growth rate increases with CAFI density to some maximum, $G_{max}(R)$. For deleterious CAFI, the growth rate decreases.

Candidate models for growth, G , might be:

$$G(D, R) = (G_0(R) + E D G_{max}(R)) / (1 + E D) \quad [for\ beneficial\ symbionts]$$

$$G(D, R) = G_0(R) e^{-E D} \quad [for\ deleterious\ symbionts]$$

in which D is CAFI density, and E is a measure of the per capita effect of CAFI; growth rates are written as functions of coral size, R , although we will start by assuming that coral growth (as measured by linear extension) is independent of coral size, as shown previously for *Pocillopora* (Kinzie and Sarmiento 1986) (but see Dornelas et al. 2017). Similar functions can be defined for mortality rates. We will then quantify relationships such as how the effect of CAFI, E , affects various measures of coral clustering, such as Ripley's K function or Clark-Evans' R (Pielou 1977).

Model Innovation 2: Incorporating flow and understanding the signal landscape. In our current model (Box 1), the probability that a CAFI settler will “choose” one coral over another is determined by the relative magnitudes of signals received from nearby coral patches. It is straightforward to include features such as signal production and degradation in our current model, but we do not currently consider water flow, which has two effects. First, it can cause *recruitment shadows* (e.g., Gaines et al. 1985, Gaines and Roughgarden 1987, Jones 1997, Palardy and Witman 2011), in which upstream corals receive more settlers than downstream corals because CAFI larvae are depleted as they pass over available coral habitat. Second, flow will alter the distribution of signal around the coral (e.g., as represented by the Peclet number in hydrodynamics: Lombard et al. 2013). In its simplest version, we can model the concentration of signal $C(x, t)$ as the solution to a partial differential equation that accounts for the signal's diffusivity D , degradation rate μ , a set of site-specific signal emission rates λ_i that emerge from the locations x_i , with a background fluid flow field V :

$$\frac{\partial C}{\partial t} = D \Delta C - \mu C + V \cdot \nabla C + \sum_i \lambda_i \delta(x - x_i)$$

where Δ is the Laplacian operator, $\cdot$ is the dot product, ∇ is the gradient operator, and δ is the Dirac delta function. This equation is a natural generalization of the model used in (Hein and McKinley 2012), but which includes flow. For a given landscape configuration, flow environment, and rates of coral signal production and degradation informed by empirical work (Aim 2), we will be able to produce a *signal landscape*, which together with larval sensitivity (θ , Box 1) yields the “basin of attraction” associated with each coral colony, a visualization of how coral signaling and environmental factors shape and predict CAFI settlement.

ACTIVITY 4-2. Parameter estimation and model checking

Our model's parameters can be separated into three groups: those associated with the CAFI settlement (e.g., involving propagule redirection), those associated with CAFI post-settlement processes (e.g., density-dependent mortality), and those associated with coral growth and survival (e.g., as mediated by the density of CAFI). Some parameters will be informed by existing data (e.g., for post-settlement density-dependent mortality). Others (e.g., involving signal production and degradation) will be estimated based on the empirical work proposed in Aims 2 and 3.

Bayesian parameter estimation: For a given coral spatial arrangement and flow environment, the model described in Activity 4-1 yields an expected number of settlers at individual corals. Then, allowing for stochasticity (e.g., Poisson distributed noise), we will construct a likelihood function for any experimental observation. Using Bayesian techniques, we will use this likelihood function and our experimental observations to produce a posterior distribution on parameter space. Sampling from this posterior distribution, we will establish a 95% credible region in parameter space and, using marginals of the posterior distribution, we will assess which of the parameters are well-specified by the data. For those that are not well-specified, we will determine if the problem arises from unidentifiable parameter combinations to help us improve the design of future experiments. For example, a preliminary theoretical analysis shows that if we did not sample DMS (Activity 2-2) at sites (or during times) with different current speeds, then it would not be possible to separately estimate signal diffusivity and the rate of signal degradation because the solution depends on their ratio (Hein and McKinley 2012).

Model checking: Because the empirical data will be noisy and have a relatively small sample size, we cannot set aside a substantial portion of the data for predictive validation. However, we will conduct an analogue of validation known as posterior predictive model checking (Gelman et al. 2013, Hooten and Hobbs 2015) or model criticism (Box 1980). This approach establishes representative summary statistics for the data, identifies a 95% range for each statistic when simulations are generated from parameter sets sampled from the posterior distribution, and checks that the summary statistics for the empirical data fall within those ranges: i.e., checks that the parameter estimates yield plausible system behavior. Some examples of possible summary statistics in the CAFI settlement experiments are the coefficient of variation in the settler counts, the ratio of settlers to external corals to internal corals, or the rate at which CAFI density decays from upstream to downstream. A good statistic should capture an essential qualitative feature of the data while not being used directly for parameter estimation. We will also use Bayesian model comparison methods to assess which features of the model are essential for describing the available data: e.g., using Watanabe-Akaike Information Criteria (Watanabe 2010) to determine whether incorporating flow in our propagule redirection model significantly improves model performance.

ACTIVITY 4-3. Scenario analysis: effects of coral-CAFI feedbacks on the rate, extent, and spatial pattern of coral recovery

Coral reef systems are frequently disturbed and there is much interest in how (and if) they recover (Graham et al. 2011, Cheal et al. 2017). Given that CAFI can affect coral growth and survival, it follows that the ability of a coral landscape to recover from a disturbance will be determined, in part, by the effect of CAFI on coral growth and survival, by the spatial pattern of the disturbance, and by the way in which CAFI assemblages change in response to the perturbation host colonies (McKeon and Moore 2014, Bruckner et al. 2017, Shaver et al. 2018). For example, we hypothesize that habitat loss will lead to increased settlement of beneficial CAFI (via propagule redirection), which should facilitate rapid coral recovery. These effects will be greatest when surviving corals are overdispersed. In contrast, if CAFI are deleterious, then recovery will be impaired by the loss of habitat, especially when surviving corals are overdispersed. To evaluate these and other hypotheses about recovery, we will conduct *in silico* experiments to extrapolate our results on coral-CAFI feedbacks to temporal scales not accessible via field experimentation.

We will (1) sample parameter combinations from qualitatively distinct parameter regimes (Activity 4-1); (2) run the model to steady state (for each parameter set); and (3) impose coral degradation on these “equilibrium” landscapes. We envision several *in silico* experiments in which we manipulate features of the coral-CAFI system or the way in which it is perturbed: e.g., vary the strength and direction of the coral-CAFI feedbacks, the magnitude of the coral loss, and the spatial pattern of coral loss (e.g., localized vs dispersed). We will then (4) measure aspects of recovery such as the time required for the system to return to 95% of its previous coral cover. Moreover, by observing various patterns of recovery, we can develop a range of possible strategies for intervention intended to accelerate landscape recovery. For example, considerable attention has been given to the outplanting of propagated corals to restore coral reef systems, but little attention has been given to their spatial configuration or the effects that will be mediated through their CAFI (Shaver and Silliman 2017).

E. FUTURE DIRECTIONS

Our project will quantify the role of feedbacks between coral and CAFI in shaping spatial patterns and dynamics of habitat and occupants on coral reefs. Although developed and tested in coral reefs, our work should provide general insights about other habitat-occupant systems: e.g., settings in which restoration of a foundation species is assumed to restore entire ecosystems, including organisms that must colonize the habitat-forming species (e.g., Falk et al. 2010). In addition to the novelty and importance of this research from a landscape ecology perspective, this work also sets the stage for future, groundbreaking research on the evolution of signaling in coral reef systems. For example, the process of attraction to settlement habitat and the resulting competition among habitat for beneficial CAFI is very similar to concepts that underlie flower–pollinator mutualisms (Mitchell et al. 2009), which are exemplars of co-evolution. Thus, ecological feedbacks between corals and their CAFI may have led to the co-evolution of signaling systems that have not yet been fully appreciated by marine scientists. Indeed, recent studies demonstrate that many marine organisms can locate and orient to cues produced by their habitat (Simpson et al. 2005, Lecchini et al.

2005, Foretich et al. 2017). Our proposed work lays a foundation for future work that investigates signaling systems and the possible co-evolution of corals and beneficial CAFI. The implications may be starkly different, however, if CAFI have deleterious effects -- instead of a co-evolved mutualism, the feedbacks between corals and CAFI may create an arms race in which corals attempt to chemically hide from potential occupants.

F. BROADER IMPACTS

Our project has broader impacts that include direct benefits of the research to marine restoration, the training of a suite of early career scientists, and outreach to the general public.

Informing reef conservation and restoration: Marine conservation scientists and resource managers need a diversity of tools to help restore ecosystems. Recently there has been a call for techniques that vary habitat configuration and encourage natural symbioses (including CAFI) to facilitate restoration in nearshore systems (Silliman et al. 2015, Sawayama et al. 2015, Shaver and Silliman 2017). Our proposed work addresses this critical knowledge gap, and will inform how habitat restoration can be done to best facilitate recovery based on the interplay between corals and CAFI (e.g., to guide the spatial design of coral outplanting efforts). We also will facilitate the application of our research by continuing to collaborate with the NOAA Ecosystem Science Program, and apprise them of our results annually, to improve marine conservation and policy.

We will also apply our efforts to the restoration of the reef adjacent to the Gump research station. Despite being included within a marine protected area, nearby goat and pineapple farms have smothered this previously vibrant system. We will transplant the *Pocillopora* colonies collected during our field studies to the Gump station reef as “seeds” to facilitate the recovery of this degraded reef; this will be done in several configurations so that long-term responses can be compared with expectations from our model. These corals are ideal candidates for restoration in areas experiencing heavy sediment runoff because crab and shrimp occupants associated with *Pocillopora* colonies actively remove sediments from their hosts. This restoration project will also serve as a mechanism for community engagement (Forrester 1990, dela Cruz et al. 2014). We will train local Polynesian volunteers (coordinated through the Atitia Center in Moorea) to outplant corals to Gump reef and monitor the outcomes, helping enhance scientific literacy and stewardship of marine resources. The results of our activities will yield insights that will allow us to refine our recommendations for future coral outplanting programs around the globe.

Early career researcher training: This project will train two graduate students, several undergraduates, and one post-doc (see post-doc mentoring plan), as well as a technician. The graduate students will engage in an integrative program combining expertise in field ecology and mathematical biology. We will recruit students, with an eye to develop a diverse team: e.g., by working with UCSB's Bridge to Doctorate Program and UGA's Gateway to Graduate School Bridge Program. Undergraduates will help with experiments in the lab and field, and will conduct outreach to local schools, focusing on under-resourced schools and under-represented groups in STEM. For example, UC Santa Barbara is a Hispanic-Serving Institution (HSI), and we will recruit from our diverse student body as we develop our undergraduate team. Through first-hand experience in multiple labs and in the Indo-Pacific, our project will expose these early career researchers to a diversity of cultures and scientific communities. We will employ a team-based research model for mentoring: in addition to the PIs mentoring all the trainees, the post-doc will mentor graduate students, who will mentor undergraduates. All participants will present research at scientific conferences and attend a weekly seminar, where they will develop research proposals, receive communication training, and be provided with feedback on professional talks, posters, and manuscripts.

Public outreach: Public outreach will take place at local institutions and in Moorea. For example, we will participate in UCSB's REEF (Research Experience & Education Facility) program, which is an interactive marine educational facility at UCSB that provides educational tours for K-12 and community groups that serves over 10,000 visitors annually who receive a hands-on, inquiry-based experience into marine science and research. To supplement the existing REEF instruction, we will provide scientific content and activities that explain our findings and current issues in reef ecology. In Moorea, PIs and trainees will provide presentations on reef community diversity and tours for the public through the Atitia Center, which is devoted to local education and outreach. Stier and his research group will also participate in the World

Ocean's Day celebration in Santa Barbara by presenting material on reef diversity and ecology, including a virtual reality tour of intact reef habitat. Osenberg's lab will continue their involvement in the art-and-science initiative *Imagination Squared: Pathways to Resilience* (IM², imaginationssquared.org), designed to engage the public in discussions of resilience. IM² has grown from an initial collaboration between the Schools of Ecology and Art to the entire Athens community (and hopefully beyond).

G. PERSONNEL & PROJECT TIMELINE (Fig. 8)

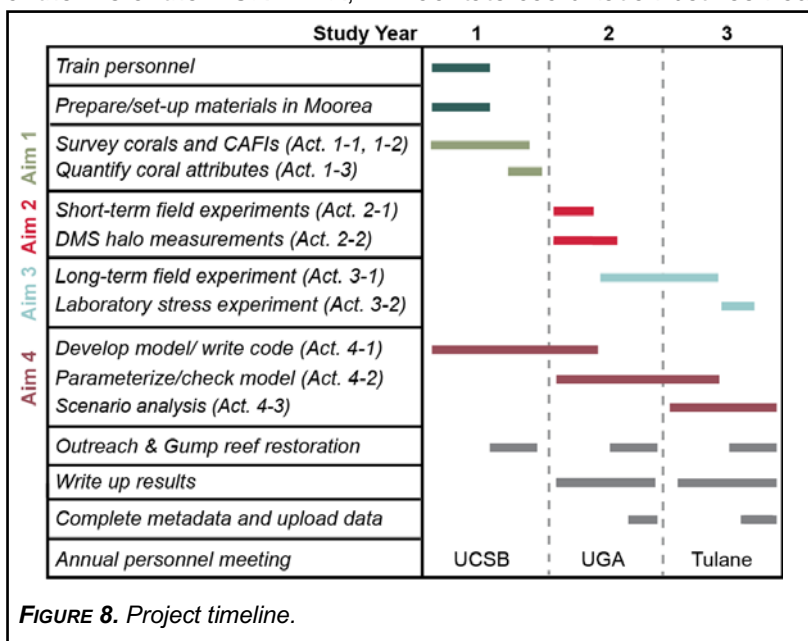
Stier's and Osenberg's labs (and the post-doc) will perform field research. Stier and Osenberg will train the graduate and undergraduate students in field and lab techniques. Schmitt provides additional expertise on coral-CAFI interactions and, as one of the PIs of the MCR-LTER, will facilitate coordination between our two programs. Stier will coordinate the field efforts in Aims 1-3 and lead the Broader Impacts (facilitated because UCSB is the home for the MCR-LTER program). Osenberg and Stier will supervise data archival. McKinley will lead the modeling efforts (Aim 4), and Osenberg will help integrate the field and modeling components.

H. RESULTS FROM PRIOR NSF SUPPORT

1. CW Osenberg, TK Frazer, JS Shima; OCE-1130359; 9/15/2011-8/31/2017; \$800,000. *Spatial patterns of coral-vermetid interactions: short-term effects and long-term consequences.*

Intellectual Merit: Highlights: (1) Established a long-term study of vermetid-coral dynamics; (2) Conducted a multi-island survey of coral-vermetid spatial patterns; (3) Investigated mechanisms of vermetid effects (e.g., via oxygen profiles, flume studies, coral microbes, allelopathy); (4) Identified a novel predator-prey interaction with vermetids; (5) Tested the local adaptation of corals and their microbial flora to vermetids; (6) Documented massive die-off of vermetids. **Broader Impacts:** (1) Facilitated the publication of 19 papers (five others in review/preparation), 3 senior theses, 6 dissertations (2 others in progress); (2) Facilitated the training of 10 PhD students (including an African-American and a Hispanic; 7 females) and 28 undergraduates; (3) Taught several workshops on meta-analysis (to ~100 participants); (4) Organized, documented, and uploaded data to BCO-DMO for public dissemination. **Publications:** see references.

2. SA McKinley, with C Hohenegger; DMS-1644290/1412998; 9/1/2014 - 8/31/2017 \$291,750 (\$125,750 UF/Tulane portion). "Collaborative Research: Diffusion of foreign particles in complex fluids" **Intellectual Merit:** (1) Extended existing mathematical theory for the movement of microparticles immersed in viscoelastic fluids, and developed novel numerical methods for simulating coupled fluid-particle systems; and (2) Challenged the existing particle tracking protocol by which fluid properties are inferred from immersed particle motion, proposed alternative methods, and provided uncertainty quantification for parameter inference. **Broader Impacts:** (1) Facilitated the publication of 7 papers (with 4 more in review or in preparation); (2) Supported 1 PhD student (graduated in May 2018), and 2 undergraduates (including a female math major); (3) Developed a course in *Stochastic Differential Equations* at Tulane. **Publications:** see references.



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- ***Bold-italic:*** Papers resulting from prior support (DMS-1644290/1412998)

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- Stier, A. C., M. A. Gil, C. S. McKeon, S. Lemer, M. Leray, S. C. Mills, and C. W. Osenberg.** 2012. Housekeeping mutualisms: Do more symbionts facilitate host performance? *PLoS ONE* 7:e32079.
- Stier, A. C., and M. Leray. 2014. Predators alter community organization of coral reef cryptofauna and reduce abundance of coral mutualists. *Coral Reefs* 33:181–191.
- Stier, A. C., C. S. McKeon, C. W. Osenberg, and J. S. Shima. 2010. Guard crabs alleviate deleterious effects of vermetid snails on a branching coral. *Coral Reefs* 29:1019–1022.
- Stier, A. C., and C. W. Osenberg. 2010. Propagule redirection: Habitat availability reduces colonization and increases recruitment in reef fishes. *Ecology* 91:2826–2832.
- Thiault, L., L. Kernaléguen, C. W. Osenberg, and J. Claudet.** 2017. Progressive-Change BACIPS: A flexible approach for environmental impact assessment. *Methods in Ecology and Evolution* 8:288–296.
- Tsounis, G., and P. J. Edmunds. 2016. The potential for self-seeding by the coral *Pocillopora* spp. in Moorea, French Polynesia. *PeerJ* 4:e2544.
- Turner, S. J. 1997. The biology and population outbreaks of the corallivorous gastropod *Drupella* on Indo-Pacific reefs. *Oceanography and Marine Biology: An Annual Review* 32:461–530.
- Veal, C. J., M. Carmi, M. Fine, and O. Hoegh-Guldberg. 2010. Increasing the accuracy of surface area estimation using single wax dipping of coral fragments. *Coral Reefs* 29:893–897.
- Wang, Y.-Y., K. L. Nunn, D. Harit, S. A. McKinley, and S. K. Lai.** 2015. Minimizing biases associated with tracking analysis of submicron particles in heterogeneous biological fluids. *Journal of Controlled Release* 220:37–43.
- Watanabe, S. 2010. Asymptotic equivalence of Bayes cross validation and widely applicable information criterion in singular learning theory. *Journal of Machine Learning Research* 11:3571–3594.
- Yeager, L. A., D. A. Keller, T. R. Burns, A. S. Pool, and F. J. Fodrie. 2016. Threshold effects of habitat fragmentation on fish diversity at landscapes scales. *Ecology* 97:2157–2166.
- Zill, J. A., M. A. Gil, and C. W. Osenberg.** 2017. When environmental factors become stressors: Interactive effects of vermetid gastropods and sedimentation on corals. *Biology Letters* 13:20160957.

ADRIAN C. STIER

Dept. Ecology Evolution & Marine Biology
UC Santa Barbara
Santa Barbara, California 93106

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a. Professional Preparation

Hamilton College	Biology (honors)	BA, 2005
U. Florida	Biology	PhD, 2012
U. British Columbia	Biodiversity Centre	Post-doc, 2012-2013
UC Santa Barbara	NCEAS	Post-doc, 2013-2016
U. Washington	SAFS	Post-doc, 2016

b. Appointments

2016 - Present Assistant Professor, EEMB, UC Santa Barbara

c. Products (all dealing with food web interactions, mutualisms, and coral reefs)

Five Publications Most Relevant to Proposed Research

Stier AC, KM Hanson, SJ Holbrook, RJ Schmitt, AJ Brooks. 2014. Predation and landscape characteristics independently affect reef fish community organization. *Ecology*. 95: 1294-1307. [PDF](#)

Stier AC and CW Osenberg. 2010. Propagule redirection: Habitat availability affects colonization and recruitment in reef fishes. *Ecology* 91: 2826–2832.

Stier AC, MA Gil, CS McKeon, S Lemer, M Leray, SC Mills, and CW Osenberg. 2012. Housekeeping mutualisms: Do more symbionts facilitate host performance. *PLoS ONE*. 7(4): e32079. [PDF](#)

Stier AC and M Leray. 2014. Predators alter community organization of coral reef cryptofauna and reduce abundance of coral mutualists. *Coral Reefs*. 33: 181-191. [PDF](#)

Stier AC, McKeon CS, Osenberg CW and JS Shima. 2010. Guard crabs alleviate deleterious effects of vermetid snails on a branching coral. *Coral Reefs*. 29 (4) 1019 - 1022. [PDF](#)

Five Additional Relevant Publications

Stier AC, Hein AM, Parravicini V, and M Kulbicki. 2014. Biogeography drives marine food web structure. *Nature Communications*. doi:10.1038/ncomms6575. [PDF](#)

White JW, Samhuri JF, **Stier AC**, Wormald CL, Hamilton SL, and SA Sandin. 2010. Synthesizing mechanisms of density dependence in reef fishes: behavior, habitat configuration, and observational scale. *Ecology*. 89: 2980–2985. [PDF](#)

Palmer TM, EG Pringle, **AC Stier**, and RD Holt. 2015. Mutualism in a community context. Book Chapter In J. Bronstein (ed) *Mutualisms*.

Geange SW and **AC Stier**. 2010. Priority effects and habitat complexity affect the strength of competition. *Oecologia*. 1: 111-118. [PDF](#)

McKeon CS, **Stier AC**, and SE McIlroy, and BM Bolker. 2012. Multiple defender effects: Synergistic coral defense by exosymbiotic crustaceans. 169: 1095-1103. *Oecologia*. [PDF](#)

d. Synergistic activities

Collaborative Research: During my PhD and post-doctoral appointments, I have held three different fellowships and two collaborative grants. Through those endeavors, I led working groups of diverse researchers including molecular biologists, ecologists, evolutionary biologists, and conservation biologists. In particular, I recently participated in national and international working groups, including the Ocean Tipping Points project, and led manuscripts in an NCEAS working group connecting biological tipping points to ocean conservation and management.

Cross disciplinary research and training programs: (1) I co-authored and accomplished the goals of a grant which was focused on understanding the importance of fundamental traits (e.g. body size, age, and life stage) in constraining vertebrate limb development. This grant included the training of 6 different undergraduate research assistants. (2) I have been a participant in an international graduate exchange program between France and the US, where I successfully conducted a series of three projects.

Undergraduate and graduate training: I have supervised and trained 10 undergraduate students and additionally served as a MS thesis co-advisor for 5 separate students at Northeastern University during a teaching appointment for a field course. To date, undergraduate and graduate students I have supervised have acted as lead author on three peer-reviewed publications. I currently am the chair of 5 graduate students (4 PhD student, 1 MS), am a committee member for four additional PhD students, and am mentoring eight undergraduates volunteering in my lab.

Professional Service (Editing, Reviewing, Advising, and Impact Assessment): My contributions to the scientific community include: (1) Acting as a scientific peer review for 20 peer-reviewed journals such as *Proceedings of the National Academy of Sciences*, *American Naturalist*, *Proceedings of the Royal Society B*, *Ecology Letters*, *Ecology*, *Nature Communications*, *Journal of Ecology*, *Oecologia*, *Oikos*, *Limnology and Oceanography*, and *BMC Ecology*, and *Coral Reefs*, and *Marine Ecology*. I am also a review editor for *Frontiers in Marine Science*. (2) Graduate mentoring (e.g. serving on yearly committees and panel discussions for graduate student-advisor relationships), and (3) Impact assessment (e.g. evaluating Marine Protected Areas in Moorea, French Polynesia).

RUSSELL J. SCHMITT

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a. Professional Preparation

University of Colorado Boulder	Biology	BS 1972
University of the Pacific	Marine Science	MS 1975
University of California Los Angeles	Biological Sciences	PhD 1979
University of California Santa Barbara	Ecology,	Post-doc, 1979-81

b. Appointments

1987-present Professor, Dept. of Ecology, Evolution and Marine Biology, University of California, Santa Barbara

1981-1987 Associate Professor, Dept. of Ecology, Evolution and Marine Biology, University of California, Santa Barbara

1975-1981 Assistant Professor, Dept. of Biological Sciences, University of California, Santa Barbara

c. Products

Five Publications Most Relevant to Proposed Research

- Lenihan, HS, JL Hensch, SJ Holbrook, **RJ Schmitt** and M Potoski. 2015. Hydrodynamics influence coral performance through simultaneous direct and indirect effects. *Ecology* 96:1540-1549.
- Stewart, HL, NN Price, SJ Holbrook, **RJ Schmitt** and AJ Brooks. 2013. Determinants of the onset and strength of mutualistic interactions between branching corals and associated crabs. *Marine Ecology Progress Series* 493 :155-163.
- Holbrook, SJ, **RJ Schmitt**, AJ Brooks. 2011. Indirect effects of species interactions on habitat provisioning. *Oecologia* 166: 739-749.
- Schmitt, RJ**, SJ Holbrook, AJ Brooks and JCP Lape. 2009. Intraguild predation in a structured habitat: distinguishing multiple predator effects from competitor effects. *Ecology* 90:2434-2443.
- Schmitt, RJ** and SJ Holbrook. 1996. Local-scale patterns of larval settlement in a planktivorous damselfish – do they predict recruitment? *Marine and Freshwater Research* 47 :449-463.

Five Additional Relevant Publications

T Guo, A Capra, M Troyer, A Gruen, AJ Brooks, JL Hench, **RJ Schmitt**, SJ Holbrook, M Dubbini. Accuracy assessment of underwater photogrammetric three-dimensional modelling for coral reefs. ISPRS — International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences. 2016; XLI-B5(June):821–828.

Messmer, V, GP Jones, PL Munday, SJ Holbrook, **RJ Schmitt** and AJ Brooks. 2011. Habitat biodiversity as a determinant of fish community structure on coral reefs. *Ecology* 92:2285-2298.

Adam, TC, **RJ Schmitt**, SJ Holbrook, AJ Brooks, PJ Edmunds, RC Carpenter and G Bernardi. 2011. Herbivory, connectivity and ecosystem resilience: response of a coral reef to a large-scale disturbance. *PLoS One* 7(9):e44953.

Schmitt, RJ and SJ Holbrook. 2007. The scale and cause of spatial heterogeneity in strength of temporal density dependence. *Ecology* 88:1241-1249.

Schmitt, RJ and SJ Holbrook. 2003. Mutualism can mediate competition and promote coexistence. *Ecology Letters* 6 :898-902.

(d) Synergistic Activities

- Lead PI of the Moorea Coral Reef LTER program (2004-present); participant in MCR LTER investigations of ecological structure and function of coral reef communities, non-linear community dynamics; member of the LTER Network Science Council
- Associate Investigator of Santa Barbara Coastal LTER program. Participant in SB LTER investigations of trophic structure and interactions on California rocky reefs
- Successfully supervised and trained 19 graduate students (12 from groups under-represented in STEM disciplines) and 6 post-doctoral scholars; mentor 2-4 REU undergraduate students per year (21 total in past 6 years); involved 6 K-12 teachers in field research (past 6 years)
- Director, Coastal Research Center of Marine Science Institute; facilitate interdisciplinary research and research training activities in coastal marine environments
- Member of the Scientific Advisory Panel to the California Coastal Commission, a 3 member group that provides scientific advice to the State of California regarding the mandated program to mitigate coastal environmental effects of San Onofre Nuclear Generating Station

CRAIG W. OSENBERG

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a. Professional Preparation

UC Santa Barbara	Biological Sci. (summa cum laude)	BA, 1980
Michigan State University	Zoology	PhD, 1988
UC Santa Barbara	Marine Sciences	Post-doc, 1988-91

b. Appointments (since 1995)

2015-life	Fellow (elected), Ecological Society of America
2014-present	Professor, Odum School of Ecology, University of Georgia
2014-2018	Courtesy Professor and Graduate Faculty, Dept. of Biology, Univ. of Florida
2013-2014	Chair, Department of Biology, University of Florida
2003-2014	Professor, Department of Biology, University of Florida
2007-2009	Honorary Research Associate, Victoria University of Wellington
2007	Visiting Professor, University of Perpignan
1995-2003	Asst. and Assoc. Professor, Department of Zoology, University of Florida

c. Products (U=undergraduate; G=graduate student; P=post-doc in or working with my lab group)

Five Publications Most Relevant to Proposed Research

- Hamman^G, EA, SA McKinley, AC Stier, **CW Osenberg**. 2018. Landscape configuration drives persistent spatial patterns of occupant distributions. *Theoretical Ecology* 11:111-127. doi:10.1007/s12080-017-0352-1
- Stier^G, AC, MA Gil^G, CS McKeon^G, S Lemer^G, M Leray^G, SC Mills, and **CW Osenberg**. 2012. Housekeeping mutualisms: do more symbionts facilitate host performance. *PLoS ONE* 7:e32079 (6 pages). doi:10.1371/journal.pone.0032079
- Stier^G, AC and CW Osenberg. 2010. Propagule redirection: Habitat availability affects colonization and recruitment in reef fishes. *Ecology* 91 : 2826–2832.
- Shima, JS, **CW Osenberg**, AC Stier^G. 2010. The vermetid gastropod *Dendropoma maximum* reduces coral growth and survival. *Biology Letters* 6:815-818.
- Osenberg, CW**, CM St. Mary, RJ Schmitt, SJ Holbrook, P Chesson, B Byrne. 2002. Rethinking ecological inference: density dependence in reef fishes. *Ecology Letters* 5:715-721.

Five Additional Relevant Publications

- Zill^U, JA, MA Gil^G, **CW Osenberg**. 2017. When environmental factors become stressors: interactive effects of vermetid gastropods and sedimentation on corals. *Biology Letters* 13. DOI: 10.1098/rsbl.2016.0957
- Shima, JS, NE Phillips, **CW Osenberg**. 2016. Variation in the growth and survival of the tropical vermetid, *Ceraesignum maximum*, is driven by size, habitat and density. *Marine Biology* 163:84. DOI 10.1007/s00227-016-2859-1

- Shima JS, NE Phillips, **CW Osenberg**. 2013. Consistent deleterious effects of vermetid gastropods on coral performance. *Journal of Experimental Marine Biology and Ecology* 439:1-6. doi:10.1016/j.jembe.2012.10.012
- Stier^G, AC, CS McKeon^G, **CW Osenberg**, JS Shima. 2010. Guard crabs alleviate deleterious effects of vermetid snails on a branching coral. *Coral Reefs* 29:1019-1022. DOI 10.1007/s00338-010-0663-9
- Claudet^P, J, **CW Osenberg** and 19 others. 2010. Marine reserves: fish life history and ecological traits matter. *Ecological Applications* 20:830-839.

d. Synergistic activities

- I have held 25 **collaborative federal grants**, 2 collaborative state funded grants, and 3 international grants, involving aquatic ecologists, theoreticians, mathematicians, statisticians, larval biologists, physical oceanographers, biogeochemists, environmental chemists, soil scientists, and developmental biologists. Subjects of these grants include coral reef ecology, stage-structured dynamics of marine and freshwater fishes, the application of meta-analysis in ecology, soil C dynamics, impact assessment designs, ecotoxicology, aquatic restoration, and the management of marine ornamental fisheries.
- I have contributed to **cross-disciplinary training and research programs**: 1) University of California Coastal Toxicology Program; I later served as chair of the External Advisory Panel to that program. 2) PI on a program that focuses on international graduate student research exchanges between the University of Florida and the University of Perpignan (France: "Ocean Bridges"). 3) PI of an NSF graduate training grant (IGERT) focused on spatial ecology and the integration of math, statistics, and biology.
- **Undergraduate, graduate, and post-doctoral training** are critical to my professional goals. I have served on over 50 Ph.D. and M.S. committees, involved >50 undergraduates in my lab's research programs, sponsored 9 post-doctoral fellows, and taught graduate workshops in the US, France, Portugal and Italy. I served as Graduate Program Coordinator at UF and on the departmental graduate committees and university graduate councils at UF and UGA.
- **Professional Service**. I served as **Editor-in-Chief of *Oecologia*** 2001-2013; Chair of the Aquatic Section of the Ecological Society of America; associate editor for *Ecology*, and *Ecological Monographs*; advisor to EMPAFISH (an EU group synthesizing data on Mediterranean MPAs); chair of the Moorea MPA Scientific Advisory Committee; and Chair of the Department of Biology (Univ. Florida). I am currently external advisor to the Coastal Waters Consortium (a research consortium of the Gulf of Mexico Research Initiative), reviews editor for *Frontiers in Marine Science*, head of the Promotion and Tenure Promotion Unit (which functions like a Department Chair for T&P) in the Odum School of Ecology. I also have organized professional meetings and symposia, including ones on stage-structured dynamics (ESA), detection of human impacts, marine biodiversity, and harvesting of marine ornamentals.
- I have been an active participant in the **National Center for Ecological Analysis and Synthesis**. I led an NCEAS working group on meta-analysis and coordinated the publication of an *Ecology* Special Feature on meta-analysis, based upon activities of the working group. I have participated in 6 different **NCEAS** working groups, including ones focused on foodwebs and coral reef ecology. I also served as a member of the NCEAS Science Advisory Board.

SCOTT A. MCKINLEY

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a. Professional Preparation

Tulane University	Mathematics	BS 1998
Ohio State University	Mathematics	MS 2002
Ohio State University	Mathematics	PhD 2006
Duke University	Mathematics, Postdoctoral Training	2006-2010
Statistical and Mathematical Sciences Institute	Postdoctoral Associate	2009-2010

b. Appointments

2016 – Present	Associate Professor, Mathematics, Tulane University
2015-2016	Assistant Professor, Mathematics, Tulane University
2010-2015	Assistant Professor, Mathematics, University of Florida
2012	Long-term visitor, Mathematical Biosciences Institute, Columbus, OH
2009-2010	Visiting Assistant Professor, Mathematics, Duke University
2009-2010	Postdoc. Associate, Statistical and Applied Mathematical Sciences Institute
2006-2009	Assistant Research Professor, Duke University

c. Products

(i) Five Most Relevant Publications (Publications on Movement Ecology or Computational Modeling of Biological Interacting Systems)

1. “Sensing and Decision-Making in Random Search.” AM Hein, **SA McKinley**. Proceedings of the National Academy of Sciences. 109 (30) 12070-12074. (2012)
2. “Sensory Information and Encounter Rates of Interacting Species.” AM Hein, **SA McKinley**. PLoS Computational Biology 9(8): e1003178. (2013)
3. “Resource-Driven Encounters and the Implications for the Spread of Disease Among Consumers” RK Borchering, SE Bellan, JM Floyd, JRC Pulliam, **SA McKinley**. J Royal Society – Interface, 14(135), 20170555. (2017)
4. “Hidden semi-Markov models reveal multiphasic movement of the endangered Florida panther” M Van de Kirk, DP Onorato, MA Criffield, BM Bolker, BC Augustine, **SA McKinley**, MK Oli. Journal of Animal Ecology. 84(2), 576-585. (2014)
5. “Modeling of Virion Collisions in Cervicovaginal Mucus Reveals Limits on Agglutination as the Protective Mechanism of Secretory Immunoglobulin A” AChen, **SA McKinley**, F Shi, S Wang, PJ Mucha, D Harit, MG Forest, and SK Lai. PLoS ONE 10(7): e0131351. (2015)

(ii) Five Other Selected Publications (Related to Mathematical and Statistical Modeling of Particle Tracking Experiments)

1. “Model comparison for single particle tracking in biological fluids.” M Lysy, NS Pillai, DB Hill, MG Forest, J Mellnik, P Vasquez and **SA McKinley**. Journal of the American Statistical Association. 111(56) (2016)
2. “Maximum Likelihood Estimation for Single Particle, Passive Microrheology Data with Drift” J Mellnik, M Lysy, P Vasquez, N Pillai, DB Hill, JA Cribb, **SA McKinley** and MG Forest. Journal of Rheology. 60, 379 (2016)
3. “A Biophysical Basis for Mucus Solids Concentration as a Candidate Biomarker for Airways Disease.” DB Hill, PA Vasquez, J Mellnik, **SA McKinley**, A Vose, F Mu, AG Henderson, SH Donaldson, NE Alexis, RC Boucher, MG Forest. PLoS ONE, 9(2), e87681. (2014)
4. “Statistical Challenges in Microrheology.” GDidier, **SA McKinley**, DB Hill, J Fricks. Journal of Time Series Analysis (33) 5 724-743. (2012)
5. “Transient Anomalous Diffusion of Tracer Particles in Soft Matter.” **SA McKinley**, L Yao and MG Forest. Journal of Rheology, 53 (6). (2009)

d. Synergistic Activities

- Organizer of the Probability and Statistics Seminar at Tulane University (2015-present) and of the Mathematical Biology Seminar at University of Florida (2010-15). Served on the Organizing Committee for the Southeast Probability Conference in Durham, NC, 2011-2015 and organized a mini-symposium at SIAM Conference on Mathematical Aspects of Material Sciences Meeting (2013)
- Undergraduate and Graduate Training; Education Outreach: Have served or am serving as primary advisor for six PhD students in Mathematics; as co-advisor for five PhD students in Biological Sciences; and as a mentor to twenty undergraduates in Mathematics and Biology. Among the PhD students, four are women; and among the undergraduates, ten were female or underrepresented minorities. I was invited to teach a graduate workshop at the Mathematical Biosciences Institute at The Ohio State University (2012) and the Lefschetz Center for Nonlinear Dynamics at Brown University (2015); sat on a Careers Panel at the Society of Mathematical Biology Summer 2017 Annual Meeting; and given two undergraduate seminars at non-research universities. I also founded and co-organized a Volunteership Workshop for high school students in Baton Rouge, LA, called “The Bayou Footprints Initiative.”
- Served on the Council for the University of Florida IGERT: Quantifying Spatial Ecology, Evolution and Environment. (2012-15).
- Peer review: Annals of Probability; Annals of Applied Statistics; American Naturalist; Biophysical Journal; Journal of Theoretical Biology; Mathematical Biosciences; Ecological Modeling; Theoretical Population Biology; Journal of Biological Physics; Discrete and Continuous Dynamical Systems; Journal of Chemical Physics; Physics Letters A

SUMMARY PROPOSAL BUDGET

YEAR 1

ORGANIZATION University of California-Santa Barbara				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Adrian C Stier				PROPOSAL NO.	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Adrian C Stier - Assistant Professor				0.00	0.00	1.00	11,676
2. Russell Schmitt - Senior Personnel				0.00	0.00	0.25	5,344
3.							
4.							
5.							
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (2) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	1.25	17,020
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.00	0.00	0.00	0
2. (1) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				12.00	0.00	0.00	40,828
3. (1) GRADUATE STUDENTS							25,169
4. (1) UNDERGRADUATE STUDENTS							6,500
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							89,517
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							45,925
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							135,442
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							7,540
2. INTERNATIONAL							17,600
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							
2. TRAVEL 0							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							15,000
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							3,000
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							41,262
TOTAL OTHER DIRECT COSTS							59,262
H. TOTAL DIRECT COSTS (A THROUGH G)							219,844
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) Modified Total Direct Costs (Rate: 26.0000, Base: 201842)							
TOTAL INDIRECT COSTS (F&A)							52,479
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							272,323
K. SMALL BUSINESS FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							272,323
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Adrian C Stier				FOR NSF USE ONLY			
ORG. REP. NAME* Reese Raffo				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

1 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 2

ORGANIZATION University of California-Santa Barbara				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Adrian C Stier				PROPOSAL NO.	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Adrian C Stier - Principal Investigator				0.00	0.00	1.00	12,345
2. Russell Schmitt - Senior Personnel				0.00	0.00	0.25	5,611
3.							
4.							
5.							
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (2) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	1.25	17,956
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.00	0.00	0.00	0
2. (1) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				6.00	0.00	0.00	20,767
3. (1) GRADUATE STUDENTS							25,924
4. (1) UNDERGRADUATE STUDENTS							7,000
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							71,647
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							34,386
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							106,033
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							7,780
2. INTERNATIONAL							15,340
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ _____ 0							
2. TRAVEL _____ 0							
3. SUBSISTENCE _____ 0							
4. OTHER _____ 0							
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							10,000
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							3,000
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							36,852
TOTAL OTHER DIRECT COSTS							49,852
H. TOTAL DIRECT COSTS (A THROUGH G)							179,005
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) Modified Total Direct Costs (Rate: 26.0000, Base: 160343)							
TOTAL INDIRECT COSTS (F&A)							41,689
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							220,694
K. SMALL BUSINESS FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							220,694
M. COST SHARING PROPOSED LEVEL \$ 0 AGREED LEVEL IF DIFFERENT \$							
PI/PI NAME Adrian C Stier				FOR NSF USE ONLY			
ORG. REP. NAME* Reese Raffo				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

2 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

1851510

SUMMARY PROPOSAL BUDGET

YEAR 3

ORGANIZATION University of California-Santa Barbara				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Adrian C Stier				PROPOSAL NO.	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Adrian C Stier - Principal Investigator				0.00	0.00	1.00	12,592
2. Russell Schmitt - Senior Personnel				0.00	0.00	0.25	5,723
3.							
4.							
5.							
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (2) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	1.25	18,315
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.00	0.00	0.00	0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0
3. (1) GRADUATE STUDENTS							26,703
4. (1) UNDERGRADUATE STUDENTS							7,500
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							52,518
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							22,331
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							74,849
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							8,030
2. INTERNATIONAL							14,491
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							
2. TRAVEL 0							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							5,000
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							3,000
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							24,112
TOTAL OTHER DIRECT COSTS							32,112
H. TOTAL DIRECT COSTS (A THROUGH G)							129,482
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)							
Modified Total Direct Costs (Rate: 26.0000, Base: 110133)							
TOTAL INDIRECT COSTS (F&A)							28,635
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							158,117
K. SMALL BUSINESS FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							158,117
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Adrian C Stier				FOR NSF USE ONLY			
ORG. REP. NAME* Reese Raffo				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

3 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

1851510

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION University of California-Santa Barbara				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Adrian C Stier				PROPOSAL NO.		DURATION (months)	
				Proposed		Granted	
AWARD NO.							
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	
				CAL	ACAD	SUMR	Funds granted by NSF (if different)
1. Adrian C Stier - Assistant Professor				0.00	0.00	3.00	36,613
2. Russell Schmitt - Senior Personnel				0.00	0.00	0.75	16,678
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (2) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	3.75	53,291
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.00	0.00	0.00	0
2. (2) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				18.00	0.00	0.00	61,595
3. (3) GRADUATE STUDENTS							77,796
4. (3) UNDERGRADUATE STUDENTS							21,000
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							213,682
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							102,642
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							316,324
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							23,350
2. INTERNATIONAL							47,431
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							
2. TRAVEL 0							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							30,000
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							9,000
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							102,226
TOTAL OTHER DIRECT COSTS							141,226
H. TOTAL DIRECT COSTS (A THROUGH G)							528,331
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)							
TOTAL INDIRECT COSTS (F&A)							122,803
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							651,134
K. SMALL BUSINESS FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							651,134
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Adrian C Stier				FOR NSF USE ONLY			
ORG. REP. NAME* Reese Raffo				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

C *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

BUDGET JUSTIFICATION

UNIVERSITY OF CALIFORNIA, SANTA BARBARA

OVERVIEW

The budget is structured to support a three-year study with field operations based from the UC Berkeley Gump Research Station on Moorea in French Polynesia. The University of California defines “year” as the fiscal year that spans from July 1 – June 30 each calendar year.

UCSB BUDGET ALLOCATIONS

A. Salaries – Senior Personnel: Senior personnel at UCSB include Lead PI Stier and Russell Schmitt. One month of summer salary per year is requested for both UCSB senior personnel. See the UGA Budget Justification and Tulane Budget Justification for salary requests for Co-PIs Osenberg and McKinley, respectively.

B. Salaries – Other Personnel: We request 1.5 years of full salary support for a laboratory assistant who will participate in the extensive field and laboratory work required for the project. Additionally, we request 49% FTE support for 1 graduate student researcher to be supervised by PI Stier during each of the proposed three years of work. The graduate student will play a critical role in facilitating field and laboratory work, while simultaneously using this project as a tool to help develop their own questions and projects for independent thesis work. We also request support to provide 500 hours per year of work for a paid undergraduate research intern. We have found that the availability of a stipend helps to recruit stronger undergraduate candidates, and that such experiences can help encourage senior students nearing a critical career juncture to stay actively involved with research.

C. Fringe Benefits: Fringe benefits are calculated based on rates set by UCSB. Please see the budget for further details of fringe benefit rates for each project personnel.

D. Equipment: Please see UGA Budget Justification for the equipment requisition included in the budget request for this project. No equipment is included in the requested budget for UCSB.

E. Travel:

- Domestic Travel We request domestic travel funds for three project personnel to travel annually between the institutions included in this proposal. These funds will cover travel to scheduled, annual personnel meeting, in addition to providing for trips to facilitate efficient collaboration, processing of samples and data, and writing generated by this project. Additionally, we request funds for one UCSB personnel to attend a single national conference each year of the project. PI Stier will also use startup funds from UCSB to facilitate travel for graduate students or other project personnel to conferences as necessary. We have adjusted the cost of travel and conference attendance for inflation by 4% in both Year 2 and Year 3 of the budget.

- Foreign Travel Costs of foreign travel to the study site in Moorea also are requested, with the amount budgeted reflecting the anticipated number of trips to be made annually between all UCSB personnel and our experience in the cost per person per trip. Airfare will cost approximately \$1500.00 per trip from LAX to Papeete RT. We have budgeted for 7 trips in Year 1 (2 by an undergraduate, 2 by a graduate student, 2 by PI Stier and one by a research technician). In years 2 and 3, only 6 trips will be made (lacking the research technician). Costs of travel within French Polynesia include costs for taxis, rental cars and ferries. We have

included field food per diem (\$15 per person per day) for the number of UCSB personnel days in Moorea each year (440 in Year 1, 350 in Year 2, and 260 in Year 3). We have adjusted the cost of international travel for inflation by 4% in both Year 2 and Year 3 of the budget.

F. Other Direct Costs:

- **Materials and Supplies** We request \$30,000 over the course of the three year grant for research supplies. These funds will cover primarily (1) acquisition and maintenance of research supplies (waterproof paper and notebooks, SCUBA gear, laboratory supplies, handheld GPS units, z-spar epoxy, cinderblocks, shipping to Moorea, etc.), (2) computing resources for PI Stier and a graduate student (computer hardware, data storage devices, and software licenses) and (3) Permitting fees and import taxes associated with working in French Polynesia. Finally, we request \$500 per year for education and outreach supplies to support our work with the Atita Center in Moorea.

- **Publication Costs** We request a modest amount to help cover fees for publication in top tier journals, including the cost of making our work open-access. Paying these fees will immediately and substantially broaden the reach of our project outputs. The amount requested will cover one open access article per year of the project.

- **Other** We include lab fees charged by Gump Station in Moorea at a projected rate of \$49 per person per day. These fees cover boat gasoline and oil, boat use, lab space, wet lab space, and housing. We included a request in the UCSB budget to cover Gump station days for all researchers (UCSB, UGA, and Tulane) because Gump is a UC Station. The number of field days described in the budget thus represents all person-days in Moorea across all three institutions. We also request funds for research-related communication costs that will facilitate work on the project. These funds will be used to set up a shared Dropbox folder for two UCSB users per year. Access to this cloud storage platform will allow efficient sharing of project related materials and enhanced quality and version control both within UCSB and with personnel at UGA and Tulane.

I. Indirect Costs:

Indirect costs for UCSB (26% MTDC) are based on the DHHS negotiated, predetermined, off-campus rate for Research Projects for the period July 1, 2010 through June 30, 2015; the rate thereafter is provisional. There is no indirect cost on equipment, graduate student tuition and fees or participant costs.

SUMMARY PROPOSAL BUDGET

YEAR 1

ORGANIZATION				FOR NSF USE ONLY					
University of Georgia Research Foundation Inc				PROPOSAL NO.		DURATION (months)			
						Proposed	Granted		
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Craig Osenberg				AWARD NO.					
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer		Funds granted by NSF (if different)	
				CAL	ACAD	SUMR			
1. Craig W Osenberg - PI				0.00	0.00	1.00	18,438		
2.									
3.									
4.									
5.									
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0		
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	1.00	18,438		
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)									
1. (0) POST DOCTORAL SCHOLARS				0.00	0.00	0.00	0		
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0		
3. (1) GRADUATE STUDENTS							29,458		
4. (0) UNDERGRADUATE STUDENTS							4,000		
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0		
6. (0) OTHER							0		
TOTAL SALARIES AND WAGES (A + B)							51,896		
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							6,251		
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							58,147		
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)									
DMS field sampler				\$ 8,000					
TOTAL EQUIPMENT							8,000		
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							8,600		
2. INTERNATIONAL							15,820		
F. PARTICIPANT SUPPORT COSTS									
1. STIPENDS \$ _____				0					
2. TRAVEL _____				0					
3. SUBSISTENCE _____				0					
4. OTHER _____				0					
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0		
G. OTHER DIRECT COSTS									
1. MATERIALS AND SUPPLIES							4,200		
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							0		
3. CONSULTANT SERVICES							0		
4. COMPUTER SERVICES							0		
5. SUBAWARDS							0		
6. OTHER							0		
TOTAL OTHER DIRECT COSTS							4,200		
H. TOTAL DIRECT COSTS (A THROUGH G)							94,767		
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)									
MTDC (Rate: 26.0000, Base: 86767)									
TOTAL INDIRECT COSTS (F&A)							22,559		
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							117,326		
K. SMALL BUSINESS FEE							0		
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							117,326		
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$					
PI/PI NAME				FOR NSF USE ONLY					
Craig Osenberg				INDIRECT COST RATE VERIFICATION					
ORG. REP. NAME*				Date Checked	Date Of Rate Sheet	Initials - ORG			
Jennifer Jones									

1 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR **2**

ORGANIZATION University of Georgia Research Foundation Inc				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Craig Osenberg				PROPOSAL NO.	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Craig W Osenberg - PI				0.00	0.00	1.00	19,175
2.							
3.							
4.							
5.							
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	1.00	19,175
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.00	0.00	0.00	0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0
3. (1) GRADUATE STUDENTS							30,636
4. (0) UNDERGRADUATE STUDENTS							4,160
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							53,971
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							6,501
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							60,472
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							4,784
2. INTERNATIONAL							16,453
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							
2. TRAVEL 0							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							4,200
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							3,000
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							0
TOTAL OTHER DIRECT COSTS							7,200
H. TOTAL DIRECT COSTS (A THROUGH G)							88,909
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 26.0000, Base: 88910)							
TOTAL INDIRECT COSTS (F&A)							23,117
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							112,026
K. SMALL BUSINESS FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							112,026
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Craig Osenberg				FOR NSF USE ONLY			
ORG. REP. NAME* Jennifer Jones				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

2 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 3

ORGANIZATION				FOR NSF USE ONLY			
University of Georgia Research Foundation Inc				PROPOSAL NO.		DURATION (months)	
						Proposed	Granted
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Craig Osenberg				AWARD NO.			
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Craig W Osenberg - PI				0.00	0.00	1.00	19,942
2.							
3.							
4.							
5.							
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	1.00	19,942
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.00	0.00	0.00	0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0
3. (1) GRADUATE STUDENTS							31,862
4. (0) UNDERGRADUATE STUDENTS							4,326
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							56,130
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							6,761
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							62,891
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							9,302
2. INTERNATIONAL							15,813
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							
2. TRAVEL 0							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							2,200
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							3,000
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							0
TOTAL OTHER DIRECT COSTS							5,200
H. TOTAL DIRECT COSTS (A THROUGH G)							93,206
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 26.0000, Base: 93206)							
TOTAL INDIRECT COSTS (F&A)							24,234
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							117,440
K. SMALL BUSINESS FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							117,440
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Craig Osenberg				FOR NSF USE ONLY			
ORG. REP. NAME* Jennifer Jones				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

3 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

1851032

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION University of Georgia Research Foundation Inc				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Craig Osenberg				PROPOSAL NO.		DURATION (months)	
				AWARD NO.		Proposed	Granted
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	
				CAL	ACAD	SUMR	Funds granted by NSF (if different)
1. Craig W Osenberg - PI				0.00	0.00	3.00	57,555
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	3.00	57,555
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.00	0.00	0.00	0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0
3. (3) GRADUATE STUDENTS							91,956
4. (0) UNDERGRADUATE STUDENTS							12,486
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							161,997
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							19,513
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							181,510
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
\$ 8,000							
TOTAL EQUIPMENT							8,000
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							22,686
2. INTERNATIONAL							48,086
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							
2. TRAVEL 0							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							10,600
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							6,000
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							0
TOTAL OTHER DIRECT COSTS							16,600
H. TOTAL DIRECT COSTS (A THROUGH G)							276,882
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)							
TOTAL INDIRECT COSTS (F&A)							69,910
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							346,792
K. SMALL BUSINESS FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							346,792
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Craig Osenberg				FOR NSF USE ONLY			
ORG. REP. NAME* Jennifer Jones				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

C *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

BUDGET JUSTIFICATION

UNIVERSITY OF GEORGIA

OVERVIEW

The budget is structured to support a three-year study with field operations based from the UC Berkeley Gump Research Station on Moorea in French Polynesia. The 3-year research program includes field surveys and field experiments, as well as an integrative modeling component.

A. Senior Personnel:

- a. Osenberg (PI) will participate in the field activities and will also advise the modeling activities. 1 month of summer salary support is requested for the PI in Years 1, 2, and 3 (1 month/yr). Osenberg will coordinate with Stier to accomplish all field-based activities in Moorea, French Polynesia.
- b. Costs are based on 2018-19 rates and inflated by 4% per year in Yrs 2 and 3.

B. Other Personnel

- a. Funds are requested for a Graduate Assistant for Years 1-3 (12 months / year at .5 FTE). The GA is essential for the extensive fieldwork that we will accomplish in each of the three years of the grant. We expect the GA to spend 6 months (in Yrs 1& 2) with the UCSB GA in the field conducting the proposed research. When not in the field, the GA will perform data QA & QC, archival, and analysis.
- b. Funds are requested for 1 undergraduate student worker in years 1-3. This student will be paid \$10/hr for approximately 400 hours/year to assist with the field experiments, DMS measurements, CAFI enumeration, and data entry.
- c. Costs are based on 2018-19 rates and inflated by 4% per year in Yrs 2 and 3.

C. Fringe Benefits

- a. Fringe benefits are estimated at 22.72% for faculty summer salary, and 7% for the graduate assistant. Undergraduate students do not carry any fringe benefits.
- b. Fringe benefits are charged at actual rates as the expenses accrue.

D. Equipment

- a. Funds are requested in Yr 1 to purchase a DMS field sampler (\$8000), which will be used to characterize signal propagation around corals and its distribution throughout the water column. The sampler is based upon the Vapor Generation-chemiluminescence (VG-CL) design of Nagahata et al. (2013) and will be fabricated by SICEM Services of Athens, GA.

E. Travel

a. Domestic:

- i. Funds are requested to support 1 trip/year for Osenberg and the GA to attend a scientific conference (location/conference TBD) to present project results. An estimated flat rate for conference attendance is \$2,300/person in year 1 with an estimated inflation rate of 4% in Yrs 2 and 3 to account for potential increases in travel costs. Actual travel costs will be reimbursed according to GSA rates in place at the time of travel.

- ii. Funds are requested for Osenberg and GA to attend annual project personnel meetings. Meetings will rotate annually between collaborating institutions. We have budgeted a flat rate of \$2,000 in Yr 1 with an estimated rate of inflation of 4%. In Yr 2, we plan to meet at UGA, so no funds are requested in Yr 2.
- iii. All domestic travel was estimated based on per traveler expenses of \$550 (per airline ticket), \$100 shuttle (to and from ATL airport), \$50 cab fare at destination, \$1000 (lodging), \$300 (per diem for meals), and \$300 (for conference registration, when appropriate).

b. Foreign:

- i. We request funds to support two trips/year for Osenberg, one trip/year for the GA, and one trip per year for an undergraduate assistant to conduct research in French Polynesia.
- ii. We estimated per traveler expenses assuming a total for shuttles to ATL, roundtrip airfare from ATL-LAX-PPT, overnight lodging in Papeete (while we are in transit to and from Moorea), cabs and ferries within French Polynesia at \$2800/trip in Yr 1; all inflated by 4%/year). We also request a modest amount of per diem to cover food expenses while in French Polynesia (\$15/person-day: 308 person-days in Yrs 1&2; 228 person-days in Year 3).
- iii. Note that lodging expenses in Moorea are included in the lab fees that will be paid through the UCSB portion of the project.

G. Other Direct Costs

- a. **Materials and Supplies.** We request a total of \$10,600 in materials and supplies (\$4200 in each of Yrs 1&2 and \$2200 in Yr 3) to purchase field supplies and gear (e.g., scuba gear, cameras, GPS, underwater slates and paper, pencils, anesthetic, squirt bottles, nets, sample jars, chemicals), to service scuba equipment, and to support Broader Impacts.
- b. **Publication costs.** We request a total of \$6000 (dispersed evenly over Years 2-3) to defray the cost of publishing our results in major ecological journals (e.g., in *Ecology* or *PLoS*), and base this amount on annual costs that we have incurred in our previous projects.
- c. **Other.** We also request funds for two Dropbox Plus licenses per year. Access to this storage platform will allow security, back-up, and efficient sharing of project-related materials both within UGA and with personnel at UCSB and Tulane.

H. Indirect Costs. IDC is calculated at the federally negotiated rate of 26% MTDC for off-campus research projects per the F&A Rate Agreement negotiated with the Department of Health and Human Services dated 5/30/2018.

SUMMARY PROPOSAL BUDGET

YEAR 1

ORGANIZATION Tulane University				FOR NSF USE ONLY					
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Scott McKinley				PROPOSAL NO.		DURATION (months)			
				Proposed		Granted			
AWARD NO.									
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer		Funds granted by NSF (if different)	
				CAL	ACAD	SUMR			
1. Scott A McKinley - Associate Professor				0.00	0.00	1.00	11,864		
2.									
3.									
4.									
5.									
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0		
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	1.00	11,864		
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)									
1. (1) POST DOCTORAL SCHOLARS				12.00	0.00	0.00	55,000		
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0		
3. (0) GRADUATE STUDENTS							0		
4. (0) UNDERGRADUATE STUDENTS							0		
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0		
6. (0) OTHER							0		
TOTAL SALARIES AND WAGES (A + B)							66,864		
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							9,570		
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							76,434		
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)									
TOTAL EQUIPMENT							0		
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							6,000		
2. INTERNATIONAL							3,000		
F. PARTICIPANT SUPPORT COSTS									
1. STIPENDS \$ _____				0					
2. TRAVEL _____				0					
3. SUBSISTENCE _____				0					
4. OTHER _____				0					
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0		
G. OTHER DIRECT COSTS									
1. MATERIALS AND SUPPLIES							0		
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							0		
3. CONSULTANT SERVICES							0		
4. COMPUTER SERVICES							0		
5. SUBAWARDS							0		
6. OTHER							0		
TOTAL OTHER DIRECT COSTS							0		
H. TOTAL DIRECT COSTS (A THROUGH G)							85,434		
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)									
Facilities and Administration (Rate: 51.5000, Base: 85434)									
TOTAL INDIRECT COSTS (F&A)							43,999		
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							129,433		
K. SMALL BUSINESS FEE							0		
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							129,433		
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$					
PI/PI NAME Scott McKinley				FOR NSF USE ONLY					
ORG. REP. NAME*				INDIRECT COST RATE VERIFICATION					
Norey Laug				Date Checked		Date Of Rate Sheet		Initials - ORG	

1 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR **2**

ORGANIZATION Tulane University				FOR NSF USE ONLY					
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Scott McKinley				PROPOSAL NO.		DURATION (months)			
				Proposed		Granted			
AWARD NO.									
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer		Funds granted by NSF (if different)	
				CAL	ACAD	SUMR			
1. Scott A McKinley - Associate Professor				0.00	0.00	1.00	12,220		
2.									
3.									
4.									
5.									
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0		
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	1.00	12,220		
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)									
1. (1) POST DOCTORAL SCHOLARS				12.00	0.00	0.00	56,650		
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0		
3. (0) GRADUATE STUDENTS							0		
4. (0) UNDERGRADUATE STUDENTS							0		
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0		
6. (0) OTHER							0		
TOTAL SALARIES AND WAGES (A + B)							68,870		
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							9,857		
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							78,727		
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)									
TOTAL EQUIPMENT							0		
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							6,000		
2. INTERNATIONAL							3,000		
F. PARTICIPANT SUPPORT COSTS									
1. STIPENDS \$ 0									
2. TRAVEL 0									
3. SUBSISTENCE 0									
4. OTHER 0									
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0		
G. OTHER DIRECT COSTS									
1. MATERIALS AND SUPPLIES							0		
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							3,000		
3. CONSULTANT SERVICES							0		
4. COMPUTER SERVICES							0		
5. SUBAWARDS							0		
6. OTHER							0		
TOTAL OTHER DIRECT COSTS							3,000		
H. TOTAL DIRECT COSTS (A THROUGH G)							90,727		
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) Facilities and Administration (Rate: 51.5000, Base: 90727)									
TOTAL INDIRECT COSTS (F&A)							46,724		
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							137,451		
K. SMALL BUSINESS FEE							0		
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							137,451		
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$					
PI/PI NAME Scott McKinley				FOR NSF USE ONLY					
ORG. REP. NAME* Norey Laug				INDIRECT COST RATE VERIFICATION					
				Date Checked		Date Of Rate Sheet		Initials - ORG	

2 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION Tulane University				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Scott McKinley				PROPOSAL NO.	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Scott A McKinley - Associate Professor				0.00	0.00	3.00	36,671
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	3.00	36,671
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (3) POST DOCTORAL SCHOLARS				36.00	0.00	0.00	170,000
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0
3. (0) GRADUATE STUDENTS							0
4. (0) UNDERGRADUATE STUDENTS							0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							206,671
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							29,580
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							236,251
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							18,000
2. INTERNATIONAL							6,000
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							
2. TRAVEL 0							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							0
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							6,000
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							0
TOTAL OTHER DIRECT COSTS							6,000
H. TOTAL DIRECT COSTS (A THROUGH G)							266,251
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)							
TOTAL INDIRECT COSTS (F&A)							137,119
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							403,370
K. SMALL BUSINESS FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							403,370
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Scott McKinley				FOR NSF USE ONLY			
ORG. REP. NAME* Norey Laug				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

C *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

BUDGET JUSTIFICATION

TULANE UNIVERSITY

Personnel

Principal Investigator: Scott McKinley. The PI will conduct this research at Tulane University in New Orleans, LA. The calculated financial support corresponds to one month of summer salary in each year of the grant with three percent annual increase.

Postdoctoral Assistant: The postdoctoral assistant will be hosted at Tulane for all three years of the grant. This individual will participate in the development and analysis of theoretical models for data produced in the early phase of the project. We request a salary of \$55,000.

Fringe Benefits

Fringe benefits are computed using the rates of 5.1% applicable to Faculty Salary during the summer, 16.3% applicable to Postdoc Salary during the academic year, 3.9% applicable to Graduate Student salary, and \$1337 for Graduate Student health insurance for the fiscal year July 1, 2018, through June 30, 2019. The graduate student fringe includes funding for health insurance. If this proposal is funded, the rates quoted above shall, at the time of funding, be subject to adjustment for any period subsequent to June 30, 2019. Tulane's fringe rates are approved annually by U.S. Department of Health and Human Services.

Travel

Domestic The PI requests \$6000 for domestic travel in each year of the grant. These funds will be used to attend at least one conference each year and to sponsor travel for the PI and the postdoc to visit collaborator institutions for periods of concentrated collaborative effort.

Foreign The PI requests \$3000 in years one and two for international travel in the second year of the grant in order to participate in research in French Polynesia. This will cover round trip airfare (MSY-LAX-PPT), ferries and taxis, and 60 days of per-diem at \$15/day.

Publication Charges

The PI expects to publish in major Ecology and Mathematics journals that support open access policies. Such journals often require substantial page charges. Based on recent experience, the PI requests \$3000 for publication charges in years 2 and 3 of the grant.

F&A – On Campus Research

The facilities and administrative costs have been calculated at Tulane's federally negotiated rate of 51.5% of the MTDC.

(See PAPPG Section II.C.2.h for guidance on information to include on this form.)

Investigator: Adrian Stier

Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Source of Support: National Science Foundation NSF

Location of Project: Off Campus

Support: ☒ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Source of Support: Society For Conservation Biology

Location of Project: UC Santa Barbara

Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Source of Support: National Science Foundation NSF

Location of Project: Off Campus

Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Source of Support: UC Multicampus Research Programs and Initiatives

Location of Project: UC Santa Barbara

Support: ☐ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Source of Support:

Location of Project:

USE ADDITIONAL SHEETS AS NECESSARY

Current and Pending Support

(See PAPPG Section II.C.2.h for guidance on information to include on this form.)

The following information should be provided for each investigator and other senior personnel. Failure to provide this information may delay consideration of this proposal.	
Investigator: Russell Schmitt	Other agencies (including NSF) to which this proposal has been/will be submitted.

Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title: COLLABORATIVE RESEARCH: Dynamic Marine Landscapes: Feedbacks and spatial patterns of coral and their associated fishes- THIS PROPOSAL	Source of Support: National Science Foundation NSF Total Award Amount: \$ 651,134 Total Award Period Covered: 05/01/19 - 04/30/22 Location of Project: Off Campus Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.00 Sumr: 0.25
Support: ☒ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title: LTER: MCR III: Long Term Dynamics of a Coral Reef Ecosystem	Source of Support: National Science Foundation NSF Total Award Amount: \$ 3,381,000 Total Award Period Covered: 09/01/16 - 08/31/22 Location of Project: Off Campus Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.90 Sumr: 1.00
Support: ☒ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title: LTER: MCR IIB: Long Term Dynamics of a Coral Reef Ecosystem	Source of Support: National Science Foundation NSF Total Award Amount: \$ 4,432,747 Total Award Period Covered: 09/01/12 - 08/31/18 Location of Project: Off Campus Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.09 Sumr: 0.00
Support: ☐ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title:	Source of Support: Total Award Amount: \$ Total Award Period Covered: Location of Project: Person-Months Per Year Committed to the Project. Cal: Acad: Sumr:
Support: ☐ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title:	Source of Support: Total Award Amount: \$ Total Award Period Covered: Location of Project: Person-Months Per Year Committed to the Project. Cal: Acad: Summ:

*If this project has previously been funded by another agency, please list and furnish information for immediately preceding funding period.

Current and Pending Support

(See PAPPG Section II.C.2.h for guidance on information to include on this form.)

The following information should be provided for each investigator and other senior personnel. Failure to provide this information may delay consideration of this proposal.	
Investigator: Craig Osenberg	Other agencies (including NSF) to which this proposal has been/will be submitted.

Support: ☒ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title: Collaborative Research: Meta-analysis: evaluation and improvement of an important synthetic tool
Source of Support: National Science Foundation Total Award Amount: \$ 221,024 Total Award Period Covered: 08/15/17 - 07/31/20 Location of Project: University of Georgia, Athens, GA Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.00 Sumr: 1.00

Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title: LTER: Georgia Coastal Ecosystems-IV
Source of Support: National Science Foundation Total Award Amount: \$ 6,762,000 Total Award Period Covered: 12/01/18 - 11/30/24 Location of Project: University of Georgia, Athens, GA Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.00 Sumr: 0.00

Support: ☒ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title: Born at the right time? Disentangling the effects of birthdate and developmental trajectories on fitness, population dynamics, and the evolution of life-history
Source of Support: Marsden Fund of Royal Society of New Zealand (all \$ to VUW) Total Award Amount: \$ 840,000 Total Award Period Covered: 01/01/16 - 12/31/19 Location of Project: Moorea, French Polynesia and Victoria University, Wellington Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.00 Sumr: 0.00

Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title: Collaborative Research: Dynamic marine landscapes -- feedbacks and spatial patterns of corals and coral-associated fishes and invertebrates
Source of Support: National Science Foundation Total Award Amount: \$ 346,792 Total Award Period Covered: 05/01/19 - 04/30/22 Location of Project: Moorea, French Polynesia and University of Georgia Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.00 Sumr: 1.00

Support: ☐ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support Project/Proposal Title:
Source of Support: Total Award Amount: \$ Total Award Period Covered: Location of Project: Person-Months Per Year Committed to the Project. Cal: Acad: Summ:

*If this project has previously been funded by another agency, please list and furnish information for immediately preceding funding period.

(See PAPPG Section II.C.2.h for guidance on information to include on this form.)

Investigator: Scott McKinley

Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Source of Support: NSF - OCE
Total Award Amount: \$ 403,370 Total Award Period Covered: 05/01/19 - 04/30/22
Location of Project: Tulane University
Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.00 Sumr: 1.00

Support: ☒ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Source of Support: NIH
Total Award Amount: \$ 297,642 Total Award Period Covered: 09/01/16 - 08/31/20
Location of Project: Tulane University
Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.00 Sumr: 1.00

Support: ☒ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Source of Support: Simons Foundation
Total Award Amount: \$ 543,126 Total Award Period Covered: 07/01/18 - 06/30/23
Location of Project: Georgia Tech University (subaward: Tulane University)
Person-Months Per Year Committed to the Project. Cal:0.00 Acad:0.00 Sumr: 0.50

Support: ☐ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Project/Proposal Title:

Source of Support:
Total Award Amount: \$ Total Award Period Covered:
Location of Project:
Person-Months Per Year Committed to the Project. Cal: Acad: Sumr:

Support: ☐ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Project/Proposal Title:

Source of Support:
Total Award Amount: \$ Total Award Period Covered:
Location of Project:
Person-Months Per Year Committed to the Project. Cal: Acad: Summ:

USE ADDITIONAL SHEETS AS NECESSARY

FACILITIES, EQUIPMENT AND OTHER RESOURCES

UNIVERSITY OF CALIFORNIA, SANTA BARBARA

Laboratory:

PI Stier has designated laboratory space (~1,000 sq. ft.) for himself as well all lab members. This includes workbenches, laboratory supplies, computing space, and a separate area with whiteboards and a conference table.

Clinical:

Not required for this project.

Animal:

Aquarium facilities at the Gump Research Station in Moorea (detailed below) provide the ability to house and maintain animals associated with our proposal. Materials and supplies available on-site will allow for the proper care, release, and (when necessary) euthanasia of study subjects following standard IACUC protocols and permitting allowances.

Computer:

Internet service is provided to all users of the University of California Berkeley Gump Research Station on its wireless network. Investigators on this project maintain computing capabilities at their respective institutions (UCSB, UGA, Tulane) commensurate with their specific research activities and travel to the field with wireless equipped laptop PCs.

PI Stier will perform data management for his portion of the project using the computing capabilities of the Marine Science Institute (MSI) at UC Santa Barbara. MSI has a 1000Mb/s connection to the UCSB campus backbone, which provides shared access to a 622Mb/s CALREN-2 connection, which in turn provides access to the Internet. MSI supports and maintains the research servers. The main data server providing network file sharing (Samba and NFS) is a running RedHat Enterprise Linux 5 (64-bit). The data server also runs SVN for revision control systems, SAS, Matlab, GSLIB and PERL for scientific applications. Currently there is 11.5 TB of storage (expandable) available on that system. The second server is running RedHat Enterprise Linux 5 (64-bit), which runs the Apache web server, and the Tomcat java servlet engine. A third server running RedHat Enterprise Linux 4 (64-bit) is the primary database server, running PostgreSQL, MySQL and the personnel database (LDAP). The Server room is connected to E-Power, and redundant power is provided by an APC 6000 UPS battery backup. Distributed server backups (via Amanda) are coordinated with MSI.

The Life Sciences Computing Group provides IT support for all biological science faculty and students at UCSB, including PI Stier's department (EEMB) and MSI.

Office:

In addition to office space and equipment provided to visiting researchers at the Gump Research Station, we have access to a telephone/fax machine, photocopier, dry work space, internet connectivity, and a printer. An additional 240 sq. ft. of dedicated office space is available to us on an as-needed basis through the MCR. This space includes a printer, flatbed scanner, two general use computers and internet connectivity. All investigators have adequate office space provided at their home institutions to meet their needs and those of the students and post-doctoral researchers associated with this project. All offices are equipped with phone and internet services. Facilities at UCSB also include a conference room with whiteboards and a large monitor, which can be used for presentations of data generated by this project or teleconferencing with other PIs.

Other:

All field work will be conducted at the UC Berkeley Gump Research Station (<http://moorea.berkeley.edu/>) on the island of Moorea in French Polynesia. The Gump Station, which is administered by the University of California Berkeley, has all of the equipment and facilities needed to support extensive field research, including laboratories and air conditioned office space. A multi-use laboratory containing a large teaching lab (~25 students), smaller research labs, a molecular lab, library, meeting room, collection and visualization lab, IT center, chemical and storage rooms, and office space was completed in 2008. Aquarium facilities include water tables with flow through seawater and space for experimental set-ups, as well as three outdoor seawater tanks. A small fleet of vehicles is available for general transportation, and 7 small water craft are available for transportation to field sites. Station housing options include a dormitory building, and seven hillside bungalows that can each accommodate 4 visitors. In addition to facilities, the Station has a permanent on-site staff of 7, which consists of a director, facilities manager, book-keeper/administrative assistant, an outreach coordinator/liaison with the Territorial Government, two maintenance persons, and a housekeeper. The University of California and the Territorial Government of French Polynesia have a formal cooperative agreement that enables scientists to conduct research at the Gump Station. Scientists must hold a research permit issued by the Territorial Government, and the Gump Station handles the paperwork and facilitates the process of obtaining research permits.

The Atitia Center is the public outreach unit of the Gump Station, and it is allied with the NGO Association Te Pu Atitia (<http://moorea.berkeley.edu/outreach/atitia/>). A major goal of the Atitia Center, through the Te Pu Atitia, is to pursue common educational and outreach programs focused on traditional knowledge, culture, and the relationship between human societies and natural ecosystems. Ms. Hinano Garnier Murphy, the Director of the Center, has expressed excitement about the contributions the Center could make to our research that explores the ecology of the reef. The Center provides us with an appropriate forum for outreach to classrooms in Moorea (since local regulations preclude our direct involvement in classrooms), as well as to the broader public through community activities and educational presentations. It also will facilitate our research because it will provide us with direct linkages to the local community; many of these individuals participate in the Association Te Pu Atitia.

FACILITIES, EQUIPMENT AND OTHER RESOURCES

UNIVERSITY OF GEORGIA

General: Osenberg's laboratory and office space at UGA is assigned in the Ecology Building and includes offices for personnel, computer workspace, and a conference facility (conference table, white boards, blackboard, large-screen monitor, and teleconferencing system). His space and available resources at UGA are sufficient to easily conduct the proposed work when personnel are on campus. As a land-grant and sea-grant institution, UGA has extensive library collections and a wide diversity of faculty in both basic and applied scientific disciplines.

Laboratory & Office: Osenberg's assigned space at UGA includes a faculty office, 3 other offices that each house 2-4 students/post-docs, a small common area with two work stations and microscopes/cameras, and a conference room for group meetings and workshops (he also shares wet lab space with colleagues). The conference room has a ShareLink Wireless Collaboration system and together with Zoom conferencing software, Osenberg can easily facilitate internal and remote meetings (including sharing desktops and documents). Additionally, OSE cultivates a very collegial approach to shared lab use in case unforeseen needs arise that cannot otherwise be satisfied within Osenberg's assigned space.

Computers and Computational Support: We have computing facilities to conduct data analyses. Osenberg's lab maintains two work-stations with licensed and freeware versions of specialized software (e.g., MATLAB, Maple, Mathematica, R, OpenBugs, OpenMEE, SAS, SigmaPlot, and ESRI products such as ArcGIS). The OSE has two full-time computer specialists who are available to assist with any computing problems; a more comprehensive UGA IT support group is available outside of the OSE. For example, the Georgia Advanced Computing Resource Center (GACRC) has a fulltime staff of six Systems Administrators and Scientific Computing Consultants who specialize in Linux/UNIX system administration, storage administration, and scientific computing consultation. GACRC computational and storage resources are available free of charge.

Other: Most of the work on this grant will be conducted in Moorea, French Polynesia, out of the Gump Research Station (<http://moorea.berkeley.edu/>) administered by the University of California Berkeley. The Gump Station has all of the equipment and facilities needed to support extensive field research, including laboratories, air conditioned office space, a molecular lab, library, meeting room, collection and visualization lab, IT center, chemical and storage rooms, water tables and aquaria with flow through seawater and space for experimental set-ups, and ~12 large outdoor seawater tanks. A small fleet of vehicles is available for terrestrial transportation, and small watercraft are available for transportation to field sites. Osenberg also has several inflatable boats, engines, and microscopes that are kept at the Gump lab and that were purchased from prior NSF grants or his university funds. These items will be available for use when existing Gump resources are not available. Station housing is available in a dormitory building, four ocean-side apartments, and seven hillside bungalows. The Station has a permanent on-site staff (director, facilities manager, book-keeper/administrative assistant, an outreach coordinator/liaison with the Territorial Government, two maintenance persons, and a housekeeper). The Gump Station handles the paperwork and facilitates the process of obtaining research permits.

FACILITIES, EQUIPMENT AND OTHER RESOURCES

TULANE UNIVERSITY, NEW ORLEANS, LA

Scientific Environment Contributing to Success

At Tulane University, the PI will mentor a graduate student and postdoctoral fellow in probability and applied math with a heavy emphasis on developing and analyzing mathematical models. In addition, we will have regular Skype meetings with our collaborators at the other sites.

Laboratory

The PI does not have lab space; however, there are several seminar rooms in Gibson Hall that are available for collaborative work.

Clinical:

No clinical studies are proposed at Tulane University.

Animal:

No animal studies are proposed at Tulane University.

Computer:

The PI has both a laptop and desktop computer available for use on this project and graduate students have access to Windows workstations in their offices. These are available both in the lab and via remote login. In addition, the PI and graduate students have access to a number of Unix clusters should the need arise for exceptional computing capacity.

Office:

The PI has an office in Gibson Hall on the campus of Tulane University and the postdoctoral fellow will also be provided an office. Graduate students have common offices in the same building.

Other:

In close proximity to the PI's office is the Mathematics Library, providing easy access to research materials. Remote access to leading journals is available via the Howard-Tilton Library website.

DATA MANAGEMENT PLAN

Data and information management for this project will be performed in close collaboration with the Moorea Coral Reef (MCR) LTER program at UC Santa Barbara, which has agreed to provide data documentation as well as storage and web access services to this project (facilitated by involvement of MCR LTER PI Schmitt). The MCR LTER Information Management System (IMS) will facilitate the archival cataloging of our data for long term preservation, enable the discovery of our data, and enhance their suitability for synthesis by us and others.

The MCR Information Management System meets or exceeds all current LTER criteria for Information Management as set forth by the LTER Information Managers Committee and LTER Network Information Systems Advisory Committee. With respect to data documentation, metadata documents in EML 2.1 dynamically derived from Metabase will be produced and maintained for our data (see below), which will make them readily adaptable to future enhancements in metadata documentation. Data accessibility will be through the MCR LTER website (and BCO-DMO) which meets or exceeds the Guidelines for LTER Website Design and Content. The website uses hierarchical navigation to provide access to the data catalog, publications and research foci.

Types of Data, Samples, and Other Materials: Multiple data types will be generated by the PIs and participants in this project including: 1) Ecological data such as species abundance, diversity, and composition of fishes and invertebrates in experimental corals stored as jpg, csv, and txt; 2) Coral condition data including coral size and growth, lipid, protein and carbohydrate content, tissue biomass, *Symbiodinium* density, chlorophyll content stored as jpg, csv, and txt; 3) Photographs of reef habitat stored as jpg, and 4) Simulation data from modeling efforts including reproducible R scripts and data outputs stored as pdf, csv, and txt.

Data and Metadata Standards: Data packages will conform to the most recent version (V2.1.0) of "EML Best Practices for LTER Sites" (August 2011). Metadata features will include embedded or online links to methods and protocols, full temporal, spatial, and taxonomic coverage, keywords from the MCR vocabulary, the NBI thesaurus, and/or the LTER Controlled Vocabulary, and units registered in the LTER Unit Dictionary. All data tables will be subject to EML Congruency Checker. Beyond the required elements, some datasets may provide explicit indexing keys and table-joining keys to facilitate cross-dataset synthesis.

Policies for Data Access, Distribution, Re-Use, Re-Distribution, and Production of Derivatives: Our databases and model codes will be subjected to the same data access policy and data distribution schema as the MCR. MCR data use policy and data release policy conform to LTER Network policies and use "Type I versus Type II" terminology. With some exceptions that are consistent with LTER data policies, MCR data are Type I (publicly available). There is no delay in releasing MCR data to the public once Quality Assurance has been verified. In accordance with LTER policy, graduate student thesis data are archived and cataloged as Type II (not released until thesis publication). Physical samples will be made available upon request where not consumed by analyses.

Policies for access and sharing will include provisions for appropriate protection of privacy, confidentiality, security, intellectual property, or other rights or requirements. To ensure accuracy and data tracking, the project will have a specific data use and re-use policy including:

- User requests require current and valid contact information that will be used by the PI for tracking and documenting data usage.
- Users are required to cite the project publications and acknowledge the NSF as the original funding source.

- Users have the final responsibility for any errors in their external and secondary analyses, while the PI and project participants will conduct quality control on the primary data and ensure accuracy of the primary data to the best of their abilities.
- The PI and project participants will not release any private or confidential information to the public, and in-house databases will be password protected.
- The PI and project participants will retain intellectual property rights, except where explicitly released for publication and documentation.

Active Data and Sample Storage: UCSB's Marine Science Institute data storage and access will be provided through infrastructure and IT resources, including a web server, database server, and filesystem server with a Storage Area Network (SAN) supporting snapshots. The archived data are supported by purposeful redundancy in backup systems provides for disaster recovery with off-site copies stored in a separate location, and allows for file restoration from more frequent on-site backups. The entire data catalog inventory is cached annually as a DVD archive off campus. Upon acquisition from the facilities, data will be quality controlled and added to the project server. All data on the project server will be accessible by all persons involved in the project.

We will use computers at UCSB, UGA, and Tulane for primary storage, statistical analysis, and modeling, with backup locally and to Dropbox (for each of inter-campus file sharing). We will also keep, store, and copy all handwritten and electronic field notes. Both during fieldwork and once we have returned to our universities, we will back up all data files on the MCR, UGA, or Tulane servers to ensure against damage or loss of the laptop and/or external hard drive. All quantitative data will be permanently archived in fully documented MCR databases and other pertinent materials will have metadata documentation that also is accessible on the MCR website.

To ensure long term usefulness of project data we will work to provide metadata of the highest quality. Data reach the MCR Information Management office in different stages of maturity depending on their type. Quality control (QC) will be performed automatically within the database, flagging further QC for human inspection. The MCR Information Manager will work with us regarding naming conventions, guidance for file organization and format, and designation of space on the server to back up raw files remotely. All PIs will be responsible for ensuring data and model codes are added to the catalog in a timely fashion.

Data Preservation and Archival: Data will be archived permanently in the original data format and also in more common, non-proprietary formats (e.g., tiff, csv, txt, fasta, etc.) to facilitate future data usage. Ecological, physiological, physical, and modeling data generated by the research and associated metadata will be stored long term on the UCSB servers, and archived at the NSF funded Biological and Chemical Oceanography Data Management Office (BCO-DMO). All data sets, derived data products, and associated metadata will be submitted and archived to the indicated databases and repositories within 2 years of collection.

Participant Roles in Data Management: The PIs are responsible for supervising all data management in cooperation with the project participants. All participants are responsible for data collection, quality control, internal database management/curation, and data publication as applicable to their research responsibilities within the project. The postdoc, graduate students, and undergraduate students will be trained in and involved in data collection, quality control, data storage, and data archival process. Given that all data are open to all project participants and will be made publicly available, no data loss or confusion would occur from departure/transition of key personnel during or after the project period.

POST-DOCTORAL MENTORING PLAN

This proposal includes one post-doctoral researcher, Dr. Elizabeth Hamman, who has played a central role in the development of our model of coral-CAFI dynamics (CAFI: coral associated fishes and invertebrates) and propagule redirection (Hamman et al. 2018). She is currently completing her first post-doc at East Carolina University (Mike McCoy's lab) on size-structured predator-prey dynamics in freshwater rock pools (via NSF support to McCoy, Ben Bolker James Vonesh, and Jeremy Wojdak). Dr. Hamman received training in Osenberg's lab at both the University of Florida and the University of Georgia, where she integrated field studies of gastropod-coral interactions with dynamic models of coral-symbiont interactions.

Our project provides her an opportunity to continue her development of the coral-CAFI modeling framework and to receive direct and regular mentorship from a mathematician – she will be based at Tulane University in McKinley's lab. She also will participate in fieldwork and help mentor the graduate students while in Mo'orea. Dr. Hamman will therefore be expanding her expertise in mathematics, while maintaining her skills in field ecology. At the onset of the project, the PIs will work with Dr. Hamman to develop an Individual Development Plan (IDP) to guide the areas of mentorship below.

(1) Multidisciplinary Training in Research Skills – The McKinley, Stier and Osenberg labs have combined expertise in marine ecology, mathematics, and modeling which will provide Dr. Hamman with the resources and intellectual environment to facilitate her ongoing integration of ecological field research, statistical methods, and mathematical modeling. Dr. McKinley will mentor her in stochastic processes and Bayesian techniques for model fitting and parameter estimation. We aim to enhance her skills in mathematical ecology and therefore expand her research and teaching opportunities.

(2) Professional Development Opportunities – Dr. Hamman's current goal is a permanent (tenure-track) faculty academic position in quantitative ecology (emphasizing the integration of field data and dynamic models). To accommodate Dr. Hamman's career goals, we will implement an adaptive mentoring strategy. Our overall goals will be to strengthen not only her skills and working knowledge in quantitative marine ecology, but also to advance her abilities in manuscript preparation, grantsmanship, teaching, and outreach, especially in both ecological and mathematical contexts. We also will work with her to refine her research/teaching statements, job talks, and interviewing skills. Dr. Hamman will have the opportunity to attend postdoc professional development seminars developed by Tulane's Office of Graduate and Postdoctoral Studies. To enhance her teaching skills and experience, Dr. Hamman will be given the opportunity to present guest lectures, develop and teach modules on mathematical applications and marine ecology, and mentor undergraduate and graduate students.

(3) Leadership Training and Communication Skills – Dr. Hamman and the PIs will meet weekly (via videoconferencing) to define the specific grant activities, assign specific responsibilities, assess progress, and discussion solutions for any challenges that arise. Dr. Hamman also will participate in the PIs' laboratory meetings, both virtually at UCSB and UGA and in person at Tulane University. Members of our laboratories use these meetings to regularly present their research, giving Dr. Hamman opportunities both to receive coaching on her communication and presentation skills and to provide similar guidance to students. Dr. Hamman will gain mentorship experience while supervising and collaborating with undergraduates, the research technician, and graduate students associated with this project. She will disseminate results from this study to colleagues at regional, national, and international conferences, such as SIAM Conference on Applications of Dynamical Systems and the annual meeting of the Ecological Society of America, providing Dr. Hamman an opportunity to showcase her research, communicate with researchers in both mathematics and ecology, expand her professional network, and establish possible future collaborations.